Survival Analysis

Lecture notes, Lectures 1–10

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Lecture 1. Introduction

1 Course Overview

- Textbook: Klein & Moeschberger, *Survival Analysis: Techniques for Censored and Truncated Data*, 2nd ed. (cited below as K&M). These notes follow its Chapters 2–5, 7–9, 11 and 12.
- The emphasis is on methodology and application, with part of the underlying theory (why the methods work and how they were developed).
- Prerequisites: basic probability theory, statistical inference, and R programming.

2 What Is Survival Data?

Definition 1.1: Survival data

Survival data track the *time until an event of interest occurs*:

$$\text{Exposure (clock starts)} \xrightarrow{\text{survival time}} \text{Event (clock ends)}.$$

Examples: time from diagnosis to death; time from treatment to cancer recurrence; time to failure of a machine or component; time to staff promotion; time to divorce; time to unsubscription.

Example 1.2: Worcester Heart Attack Study (WHAS100)

Survival times following hospitalisation for a heart attack (MI). A snapshot:

ID	Female	Age (years)	Follow-up (years)	Event (death)
16	0	39	7.2	0
66	1	91	1.0	1
71	0	65	5.1	0
55	0	80	5.3	0
61	0	90	0.3	1

Question: what is the mean survival time after hospitalisation?

- *Wrong approach 1* (use only subjects with known survival times): $(1.0 + 0.3)/2 = 0.65$. This is an **underestimate**, because we ignored the people who survived longer.
- *Wrong approach 2* (pretend follow-up time = survival time): $(7.2 + 1.0 + 5.1 + 5.3 + 0.3)/5 = 3.78$. Also an **underestimate**, because IDs 16, 71, 55 actually survived *longer* than their follow-up times.

Acknowledging censoring. Writing censored times with a "+": 7.2^+, 1.0, 5.1^+, 5.3^+, 0.3. For ID 66 we know the patient survived exactly 1 year after MI; for ID 16 we only know that the patient survived *at least* 7.2 years. The data carry **incomplete (partial) information**.

Remark 1.3: Why both naive estimates are biased downward

Let T_i be the true survival time and C_i the follow-up (censoring) time, so we observe $Y_i = \min(T_i, C_i)$. Since $Y_i \le T_i$ always, $\mathrm{E}(\bar{Y}) \le \mathrm{E}(T)$ (approach 2). Approach 1 keeps only $\{T_i \le C_i\}$, i.e. it conditions on the event time being *small enough* to be seen, which preferentially

discards long survivors.

3 What Is Special About Survival Data?

- Classical mathematical statistics is built on i.i.d. data, e.g. n i.i.d. copies $T_1, \ldots, T_n$ of the survival time.
- The **incompleteness** of survival data distinguishes survival analysis from traditional methods.
- Formulating a survival problem requires three ingredients:
  (1) the **event of interest**;
  (2) the **time origin**;
  (3) the **time scale**.
- Incompleteness is realized as either **censoring** or **truncation**. Common types: right censoring (RC), left censoring (LC), interval censoring (IC), left truncation (LT), right truncation (RT).

4 Types of Censoring

4.1 Right censoring

Definition 1.4: Right censoring (RC)

For a right-censored observation, the exact event time is unknown; all we know is that the event happens *after* a certain time point: $T_i > C_i$.

Example 1.5: Time to death in a breast cancer trial

T_i = time to death from randomization (receipt of treatment); the time of randomization is the time origin. If the study ends 9 months after the patient's treatment date and the patient is still alive, T_i is right censored at 9 months: $T_i > 9$.

4.2 Left censoring

Definition 1.6: Left censoring (LC)

For a left-censored observation, all we know is that the event happened *prior to* a certain time point: $T_i < C_i$.

Example 1.7: Time to first use of marijuana

Question: "When did you first use marijuana?" Possible answers:

- the exact age (uncensored);
- "never used it" (right censored at the current age);
- "used it but cannot recall when" (left censored at the current age).

When both left and right censoring are present in the data we speak of **double censoring**.

4.3 Interval censoring

Definition 1.8: Interval censoring (IC)

For an interval-censored observation, all we know is that the event happened within an interval: $T_i \in (L_i, R_i]$.

Example 1.9: Time to tumour recurrence

After breast cancer surgery, MRI is performed every 5 months. If the tumour is absent at month 10 and detected at month 15, then $10 < T_i \leq 15$.

Remark 1.10

RC and LC are special cases of IC: right censoring at C is $T \in (C, \infty)$, left censoring at C is $T \in [0, C)$.

5 Types of Truncation

5.1 Left truncation

Definition 1.11: Left truncation (LT)

Only individuals who "survive" a sufficient time are **included** in the sample. If V_i is the entry (truncation) time, T_i is observed only if $T_i \geq V_i$, i.e. we observe the conditional distribution of T_i given $T_i \geq V_i$.

Example 1.12: Longevity of elderly residents of a retirement community

An individual must survive to a sufficient age to enter the retirement center; anyone who died earlier never enters. The data are left truncated. Ignoring LT leads to **length-biased sampling**: long lifetimes are over-represented.

5.2 Right truncation

Definition 1.13: Right truncation (RT)

Only individuals who have *experienced the event* by a specified time are included in the sample: T_i is observed only if $T_i < R_i$.

Example 1.14: Age at onset of diabetes

Diabetic patients are enrolled and asked retrospectively for their age of onset. With $R_i =$ age at the time of the study and $T_i =$ age of onset, T_i is observed conditionally on $T_i < R_i$: a subject must already have diabetes to be included.

Example 1.15: HIV incubation time (Lagakos et al., 1988)

Data on 258 adults and 37 children infected with HIV who developed AIDS by June 30, 1986. With X = age at HIV infection and R = age at developing AIDS, the incubation time $R - X$ is known only for those who were infected *and* developed AIDS within the study window. This is right truncation.

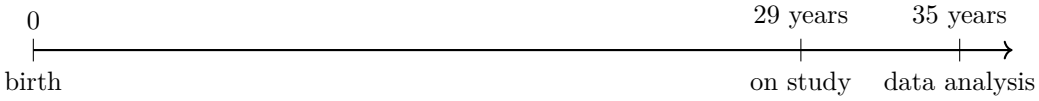
5.3 Censoring vs. truncation

Censoring	The event/subject *is detected*, but its value is not known completely: we only know it is less than and/or more than some number.
Truncation	A subject is detected *only if* its value is greater (or less) than some number; when detected, its value is completely known.

The key difference: a censored object is detectable, whereas a truncated object is not even detectable (it never enters the data set).

6 CVD Example: Time Scales, Truncation and Censoring Together

An observational study enrols subjects *without* heart disease to identify risk factors for cardiovascular disease (CVD); the event is CVD. Consider a subject who enters at age 29; data are analyzed when the subject would be 35.



Two time scales are possible depending on the origin:

- origin = birth: T_i^1 = age at CVD;
- origin = study entry: T_i^2 = years on study until CVD.

Both may be right censored (subject still CVD-free at the end of study). The first scale T_i^1 is usually more interesting.

Scenario	T_i^1 (age)	T_i^2 (time on study)
CVD at age 34	$T_i^1 = 34$ observed *given* $T_i^1 \geq 29$: **LT at 29**	$T_i^2 = 34 - 29 = 5$: no truncation, no censoring
No CVD by 35	$T_i^1 > 35$ given $T_i^1 \geq 29$: **LT at 29 and RC at 35**	$T_i^2 > 6$: **RC**, not LT
Enrolled at 30, already had CVD before 30	$T_i^1 < 30$: **LC at 30**	—

7 Definitions and Censoring Patterns

- **Failure time** (survival time, event time): the length of "time" to an event.
- **Time origin**: objectively defined and scientifically meaningful; ideally all subjects are comparable at the origin.
- **Time scale**: usually clock/calendar time (seconds, hours, weeks, months, years), but others are possible, e.g. QALY (quality-adjusted life years).

Definition 1.16: Equal or unequal censoring times

- **Type I censoring**: all subjects enter at the same real time and all censoring times are equal.
- **Generalized Type I censoring**: with staggered entry the censoring times differ, but each is known at entry.
- **Random censoring**: losses to follow-up (death from other causes, relocation, withdrawal of consent); censoring times are known only for those actually censored, and they differ.

It is crucial that the actual censoring *times* are recorded, not just the fact that a subject is censored.

8 Possible Simple (Naive) Methods

1. **Dichotomise at a pre-specified t_0.** Survivors to t_0 are "successes", deaths before t_0 "failures"; use binary-data methods (chi-square test, logistic regression).
   Disadvantages: difficulty with censoring before t_0; loss of efficiency from concentrating on a single point of the survival curve; arbitrary choice of t_0.
2. **Take logs and use the linear model.** Disadvantages: possible undue dependence on short failure times; the log-normal distribution may not be appropriate; still cannot handle censoring.

Both can be useful as a prelude to more complex analyses.

9 Summary

- Six types of incompleteness: RC, LC, IC, LT, RT (plus double censoring). LT and RC are the most common sampling schemes.
- **Censoring only**: T_i may not be observed directly; it is known to be $>$ or $<$ some value, or to lie in an interval.
- **Truncation only**: T_i is observed conditionally on $T_i >$ or $<$ some value.
- Mixtures of censoring and truncation are more complex.

10 Selected exercises

Motivation. The biases described above in words can be proved: under independence, discarding censored subjects shortens, left truncation lengthens and right truncation shortens the observed lifetimes, and the choice of time origin matters except for the exponential law.

Exercise 1.1 (Approach 1 is biased downward)

Let $T \geq 0$ with $\mathrm{E}(T) < \infty$ and let C be independent of T with $P(T \leq C) > 0$.
(a) Prove Chebyshev's association inequality: if φ is nondecreasing, g is nonincreasing, both are bounded or $\varphi(T), g(T), \varphi(T)g(T)$ are integrable, then $\mathrm{E}[\varphi(T)g(T)] \leq \mathrm{E}[\varphi(T)]\,\mathrm{E}[g(T)]$.
(b) Show that $\mathrm{E}(T \mid T \leq C) \leq \mathrm{E}(T)$.

Solution [Supplement].

(a) Let T' be an independent copy of T. For all s, t the numbers $\varphi(s) - \varphi(t)$ and $g(s) - g(t)$ have opposite signs (or one is 0), so

$$\{\varphi(T) - \varphi(T')\}\{g(T) - g(T')\} \leq 0.$$

Taking expectations and using that T, T' are i.i.d.,

$$2\mathrm{E}[\varphi(T)g(T)] - 2\mathrm{E}[\varphi(T)]\mathrm{E}[g(T)] \leq 0.$$

(b)

Step 1. Put $g(t) = P(C \geq t)$, which is nonincreasing with values in $[0, 1]$. By independence and Fubini, $P(T \leq C) = \mathrm{E}[g(T)]$ and $\mathrm{E}[T\, I(T \leq C)] = \mathrm{E}[T\, g(T)]$.

Step 2. Apply (a) with $\varphi(t) = t$ (integrable, since $\mathrm{E}T < \infty$ and $0 \leq g \leq 1$): $\mathrm{E}[Tg(T)] \leq \mathrm{E}(T)\mathrm{E}[g(T)]$.

Step 3. Dividing by $\mathrm{E}[g(T)] = P(T \leq C) > 0$,

$$\mathrm{E}(T \mid T \leq C) = \frac{\mathrm{E}[T\, I(T \leq C)]}{P(T \leq C)} \leq \mathrm{E}(T).$$

Approach 1 of the WHAS100 example estimates the left-hand side, so it is biased downward whatever the censoring distribution. ■

Exercise 1.2 (Truncation shifts the observed distribution)

Let $T \geq 0$ have survival function S.
(a) Let V be an entry time independent of T with $P(T \geq V) > 0$. Show that $P(T > t \mid T \geq V) \geq S(t)$ for every t, i.e. left-truncated lifetimes are stochastically larger than T.
(b) Let R be independent of T with $P(T < R) > 0$. Show that $P(T > t \mid T < R) \leq S(t)$ for every t.

Solution [Supplement].

(a) Let $g(s) = P(V \leq s)$, nondecreasing. By independence, $P(T \geq V) = \mathrm{E}[g(T)]$ and $P(T > t,\, T \geq V) = \mathrm{E}[I(T > t)g(T)]$. Both $s \mapsto I(s > t)$ and g are nondecreasing, so applying Exercise 1.1(a) to $\varphi = I(\cdot > t)$ and the nonincreasing $-g$ gives

$$\mathrm{E}[I(T > t)g(T)] \geq P(T > t)\,\mathrm{E}[g(T)].$$

Divide by $\mathrm{E}[g(T)] > 0$.

(b) Now $g(s) = P(R > s)$ is nonincreasing, $P(T < R) = \mathrm{E}[g(T)]$ and $P(T > t, T < R) = \mathrm{E}[I(T > t)g(T)]$. Exercise 1.1(a) with $\varphi = I(\cdot > t)$ gives $\mathrm{E}[I(T > t)g(T)] \le P(T > t)\mathrm{E}[g(T)]$; divide by $\mathrm{E}[g(T)] > 0$.

This is the precise form of length-biased sampling in the retirement-community example, and of the opposite bias in the diabetes example. ■

Exercise 1.3 (Changing the time origin)

Let T^1 (age at the event) be continuous with survival S_1 and hazard h_1, and let $a \ge 0$ with $S_1(a) > 0$. A subject enters the study at age a and is followed on the study scale $T^2 = T^1 - a$, conditionally on $T^1 \ge a$.
(a) Show that, given $T^1 \ge a$, T^2 has survival $S_1(a+u)/S_1(a)$ and hazard $h_1(a+u)$, $u \ge 0$.
(b) Assume $S_1 > 0$ on $[0, \infty)$. Show that the conditional law of T^2 equals the law of T^1 for every $a \ge 0$ if and only if T^1 is exponential.

Solution [Supplement].

(a) For $u \ge 0$, $P(T^2 > u \mid T^1 \ge a) = P(T^1 > a+u)/P(T^1 \ge a) = S_1(a+u)/S_1(a)$, using continuity. Taking $-\frac{\mathrm{d}}{\mathrm{d}u} \log$ gives the hazard $-\frac{\mathrm{d}}{\mathrm{d}u} \log S_1(a+u) = h_1(a+u)$.

(b)

Step 1. If $T^1 \sim \mathrm{Exp}(\lambda)$, then $S_1(a+u)/S_1(a) = e^{-\lambda u} = S_1(u)$.

Step 2. Conversely, suppose $S_1(a+u) = S_1(a)S_1(u)$ for all $a, u \ge 0$. Then $\psi = -\log S_1$ (finite, since $S_1 > 0$) satisfies $\psi(a+u) = \psi(a) + \psi(u)$, $\psi \ge 0$ nondecreasing.

Step 3. Additivity gives $\psi(q) = q\,\psi(1)$ for rational $q \ge 0$. For real $x \ge 0$ and rationals $q_1 \le x \le q_2$, monotonicity gives $q_1\psi(1) \le \psi(x) \le q_2\psi(1)$; letting $q_1, q_2 \to x$ yields $\psi(x) = \lambda x$ with $\lambda = \psi(1)$. Since $S_1(\infty) = 0$, $\lambda > 0$, so $T^1 \sim \mathrm{Exp}(\lambda)$.

Hence, outside the exponential model, analyzing the CVD study on the age scale (with left truncation) and on the study scale (without) answers different questions. ■

Lecture 2. Basic Quantities

1 Review: Classifying Incomplete Data

Situation	Type
Clinical trial starts at randomization (time 0); a participant completes 24 months with no event by study end.	RC
At baseline the tumour is already present; exact onset unknown, only that it occurred before baseline.	LC
Annual screening; disease-free at year 2, first positive at year 3.	IC
Registry enrols patients only if alive at clinic entry (some enter 3 years after diagnosis); followed until death or end of study.	LT + (RC or event)
Historical cohort assembled in 2020 from obituaries of people who died in 2010–2019; age at death recorded, no survivors.	RT

2 Survival Function

Let $X \geq 0$ be a random failure time.

Definition 2.1: Survival function

For continuous X with pdf f and cdf $F(x) = P(X \leq x)$,

$$S(x) = P(X > x) = 1 - F(x) = \int_x^\infty f(s)\,\mathrm{d}s.$$

$S(x)$ is the probability of surviving beyond time x. It is monotone non-increasing with $S(0) = 1$ and $S(\infty) = 0$.

Graphically, $S(x)$ is the area under the density to the right of x. (Typical examples are the Exp(0.5) density and the 1989 US all-cause survival curves by race/sex, which stay near 1 until middle age and then drop steeply.)

3 Hazard Function

Definition 2.2: Hazard function

$$h(x) = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x \mid X \geq x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x)/P(X \geq x)}{\Delta x} = \frac{f(x)}{S(x)}.$$

- $f(x)\,\mathrm{d}x \approx P(x \leq X < x + \mathrm{d}x) = S(x) - S(x + \mathrm{d}x)$ is the "infinitesimal" probability of failure at x, *unconditionally* on being alive before x.
- $h(x)\,\mathrm{d}x$ is the infinitesimal probability of failure in the next instant after x, *given* alive at x.

Remark 2.3: h is a rate, not a probability

$h(x) \geq 0$ can exceed 1; only $h(x)\Delta x$ for small Δx approximates a conditional probability.

3.1 Shapes of hazard functions

(i) **Constant**: e.g. the exponential distribution.

(ii) **Increasing**: industrial systems and cars, whose component failure rates rise over time (ageing).

(iii) **Decreasing** (less common): e.g. transplant studies (risk is highest just after the operation).

(iv) **Bathtub-shaped**: population mortality (high infant mortality, low in mid-life, rising in old age).

(v) **Hump-shaped (unimodal)**: rises early and eventually declines; e.g. breast cancer treatment studies where surgery- or toxicity-related high-risk patients tend to die early.

3.2 Examples

Example 2.4: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$ and

$$S(x) = \int_x^\infty \lambda e^{-\lambda s}\,\mathrm{d}s = \big[-e^{-\lambda s}\big]_x^\infty = e^{-\lambda x}, \qquad h(x) = \frac{f(x)}{S(x)} = \lambda.$$

The constant hazard is the *memoryless* property: $P(X > x + t \mid X > x) = e^{-\lambda t}$ does not depend on x.

Example 2.5: Weibull(α, λ)

$S(x) = \exp(-\lambda x^\alpha)$. Then

$$f(x) = -\frac{\mathrm{d}S(x)}{\mathrm{d}x} = \alpha\lambda x^{\alpha-1}\exp(-\lambda x^\alpha), \qquad h(x) = \frac{f(x)}{S(x)} = \alpha\lambda x^{\alpha-1}.$$

$\alpha > 1$: increasing hazard; $\alpha = 1$: constant (exponential); $\alpha < 1$: decreasing.

4 Cumulative Hazard and the S–h Relationship

Motivation. Survival, density, hazard and cumulative hazard are four descriptions of the same distribution, and different methods estimate different ones. The relation $S(t) = \exp\{-H(t)\}$ lets us move between them.

Theorem 2.6: Continuous case

Define the **cumulative hazard** $H(x) = \int_0^x h(s)\,\mathrm{d}s$. If X is absolutely continuous,

$$H(x) = -\ln S(x), \qquad S(x) = \exp\{-H(x)\} = \exp\Big\{-\int_0^x h(s)\,\mathrm{d}s\Big\}.$$

Proof [Supplement]. Since $f(s) = -S'(s)$,

$$H(x) = \int_0^x \frac{f(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \frac{S'(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \mathrm{d}\{\ln S(s)\} = -\big[\ln S(x) - \ln S(0)\big] = -\ln S(x),$$

using $S(0) = 1$. Exponentiating gives $S(x) = e^{-H(x)}$. ■

Summary of relations (continuous X)	
$f(t) = \dfrac{\mathrm{d}F(t)}{\mathrm{d}t} = -S'(t)$	$F(t) = 1 - S(t)$
$h(t) = \dfrac{f(t)}{S(t)} = -\dfrac{\mathrm{d}}{\mathrm{d}t}\ln S(t)$	$H(t) = \displaystyle\int_0^t h(s)\,\mathrm{d}s = -\ln S(t)$
$S(t) = \exp\{-H(t)\}$	$f(t) = h(t)\exp\{-H(t)\}$

Any one of f, F, S, h, H determines all the others.

5 Discrete Failure Times

At points $0 = x_0 < x_1 < \cdots < x_d$ let $P(X = x_j) = p_j > 0$, $\sum_{j=1}^d p_j = 1$. Discrete times arise from rounding of measurements or grouping failure times into intervals. Then

$$S(x) = P(X > x) = \sum_{j=1}^{d} P(X = x_j)\, I(x_j > x),$$

a right-continuous step function.

Definition 2.7: Discrete hazard

$$h(x_j) = P(X = x_j \mid X \geq x_j) = \frac{p_j}{S(x_{j-1})} = \frac{p_j}{\sum_{k=j}^d p_k}, \qquad j = 1, \ldots, d.$$

Here $h(x_j)$ *is* a conditional probability, so $0 < h(x_j) \leq 1$.

Since $p_j = S(x_{j-1}) - S(x_j)$,

$$h(x_j) = \frac{S(x_{j-1}) - S(x_j)}{S(x_{j-1})} = 1 - \frac{S(x_j)}{S(x_{j-1})}.$$

Example 2.8: $p_j = 1/3$, $j = 1, 2, 3$

$$S(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 2/3, & 1 \leq x < 2 \\ 1/3, & 2 \leq x < 3 \\ 0, & x \geq 3 \end{cases} \qquad h(1) = \frac{1/3}{1} = \frac{1}{3}, \quad h(2) = \frac{1/3}{2/3} = \frac{1}{2}, \quad h(3) = \frac{1/3}{1/3} = 1.$$

The last hazard equals 1: anyone alive just before the last support point must fail there.

Theorem 2.9: Product-limit representation

$$S(x_j) = \frac{S(x_j)}{S(x_{j-1})}\frac{S(x_{j-1})}{S(x_{j-2})}\cdots\frac{S(x_1)}{S(x_0)}S(x_0) = \prod_{i:x_i\le x_j}\frac{S(x_i)}{S(x_{i-1})} = \prod_{i:x_i\le x_j}[1-h(x_i)],$$

using the telescoping product and $S(x_0) = S(0) = 1$.

This product form is the population version of the Kaplan–Meier estimator (Lecture 5).

Remark 2.10: Two definitions of the discrete cumulative hazard

Klein & Moeschberger define $H(x_j) = \sum_{i:x_i\le x_j} h(x_i)$. With this definition $S(x) = \exp\{-H(x)\}$ does *not* hold. Cox & Oakes (1984) instead define

$$H(x_j) = -\sum_{i:x_i\le x_j}\log\left[1-h(x_i)\right] \approx \sum_{i:x_i\le x_j} h(x_i),$$

which restores $S = e^{-H}$ exactly (take logs of the product-limit form). The approximation uses $-\log(1-u)\approx u$ for small u.

6 Mean and Mean Residual Life

Motivation. For a patient who has survived to time t, the relevant question is the expected remaining lifetime. The mean residual life answers it, and it can be computed from the survival function alone.

Theorem 2.11: Mean as area under S

For $X\ge 0$, $\mathrm{E}(X) = \int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty S(t)\,\mathrm{d}t$.

Proof [Supplement]. Write $t = \int_0^t \mathrm{d}u$ and exchange the order of integration (Tonelli, integrand ≥ 0):

$$\int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty\int_0^t f(t)\,\mathrm{d}u\,\mathrm{d}t = \int_0^\infty\int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_0^\infty S(u)\,\mathrm{d}u.$$

■

Definition 2.12: Mean residual life

$\mathrm{mrl}(x) = \mathrm{E}(X-x\mid X>x)$. For continuous X,

$$\mathrm{mrl}(x) = \frac{\int_x^\infty (t-x)f(t)\,\mathrm{d}t}{S(x)} = \frac{\int_x^\infty S(t)\,\mathrm{d}t}{S(x)},$$

the area under the survival curve to the right of x divided by $S(x)$. Note $\mathrm{mrl}(0) = \mathrm{E}(X)$.

Proof of the second equality [Supplement]. Same Fubini argument with $t-x = \int_x^t \mathrm{d}u$: $\int_x^\infty (t-x)f(t)\,\mathrm{d}t = \int_x^\infty\int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_x^\infty S(u)\,\mathrm{d}u$. ■

7 Quantiles

Definition 2.13: p-th quantile

The p-th quantile ($100p$-th percentile) x_p satisfies $S(x_p) \le 1-p$:

$$x_p = \inf\{t : S(t) \le 1-p\}.$$

For continuous strictly decreasing S this is the unique solution of $S(x_p) = 1-p$, i.e. $F(x_p) = p$. The median is $x_{0.5}$.

Example 2.14: Exponential(λ) — mean, mrl, median

$S(x) = e^{-\lambda x}$.

- $\mathrm{E}(X) = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = 1/\lambda$.
- $\mathrm{mrl}(x) = \dfrac{\int_x^\infty e^{-\lambda t}\,\mathrm{d}t}{e^{-\lambda x}} = \dfrac{e^{-\lambda x}/\lambda}{e^{-\lambda x}} = \dfrac{1}{\lambda}$ (constant: memorylessness).
- Median: $e^{-\lambda x_{0.5}} = 1/2 \Rightarrow x_{0.5} = (\ln 2)/\lambda$.

Example 2.15: Weibull quantile

$S(x) = \exp(-\lambda x^\alpha)$, so $F(x_p) = p$ gives

$$1 - e^{-\lambda x_p^\alpha} = p \implies e^{-\lambda x_p^\alpha} = 1-p \implies \lambda x_p^\alpha = -\ln(1-p) \implies x_p = \Big[\frac{-\ln(1-p)}{\lambda}\Big]^{1/\alpha}.$$

In particular the median is $x_{0.5} = (\ln 2/\lambda)^{1/\alpha}$.

Example 2.16: LogNormal quantile

$X \sim \mathrm{LogNormal}(\mu, \sigma^2)$ means $Y = \ln X \sim N(\mu, \sigma^2)$. Since ln is strictly increasing, quantiles map through it:

$$p = P(X \le x_p) = P(Y \le \ln x_p) = \Phi\Big(\frac{\ln x_p - \mu}{\sigma}\Big) \implies x_p = \exp\left(\mu + \sigma\Phi^{-1}(p)\right),$$

where Φ^{-1} is the standard normal quantile function. With $p = 1/2$, $\Phi^{-1}(1/2) = 0$, so $\mathrm{median}(X) = e^\mu$.

8 Median Residual Life

Definition 2.17: Median residual life

Preferred to the mrl when the distribution is highly skewed.

$$\operatorname{mdrl}(x) = \operatorname{median}(X - x \mid X > x),$$

equivalently

$$P(X - x > \operatorname{mdrl}(x) \mid X > x) = \frac{1}{2} \iff \frac{P(X > x + \operatorname{mdrl}(x),\, X > x)}{P(X > x)} = \frac{1}{2}.$$

For continuous X the numerator is $S(x + \operatorname{mdrl}(x))$, hence

$$S(x + \operatorname{mdrl}(x)) = \tfrac{1}{2} S(x) \quad \Longrightarrow \quad \operatorname{mdrl}(x) = S^{-1}\Big(\frac{S(x)}{2}\Big) - x.$$

Example 2.18: Exponential(λ)

$S^{-1}(y) = -(1/\lambda)\ln y$ for $y \in (0, 1]$, so

$$\operatorname{mdrl}(x) = -\frac{1}{\lambda}\ln\Big(\frac{e^{-\lambda x}}{2}\Big) - x = -\frac{1}{\lambda}(-\lambda x - \ln 2) - x = \frac{\ln 2}{\lambda}.$$

Again constant in x.

9 Weibull Mean Residual Life

Let $S(t) = \exp(-\lambda t^{\alpha})$, $\lambda, \alpha > 0$, and $m(t) = \mathrm{E}[T - t \mid T > t]$.

1. Derive $m(t)$. 2. Show $m(t)$ is constant when $\alpha = 1$. 3. For $\alpha > 1$, show $m(t)$ is decreasing and interpret.

Derivation: substitution $v = \lambda u^{\alpha}$

With $f(u) = \lambda\alpha u^{\alpha-1} e^{-\lambda u^{\alpha}}$, the numerator is

$$N(t) = \int_t^{\infty} (u - t)\, \lambda\alpha u^{\alpha-1} e^{-\lambda u^{\alpha}}\, \mathrm{d}u.$$

Set $v = \lambda u^{\alpha}$, so $\mathrm{d}v = \lambda\alpha u^{\alpha-1}\,\mathrm{d}u$ and $u = (v/\lambda)^{1/\alpha}$; limits $u = t \mapsto v = \lambda t^{\alpha}$, $u \to \infty \mapsto v \to \infty$. Then

$$\begin{aligned} N(t) &= \int_{\lambda t^{\alpha}}^{\infty} \Big(\big(\tfrac{v}{\lambda}\big)^{1/\alpha} - t\Big) e^{-v}\,\mathrm{d}v = \lambda^{-1/\alpha} \int_{\lambda t^{\alpha}}^{\infty} v^{1/\alpha} e^{-v}\,\mathrm{d}v - t\int_{\lambda t^{\alpha}}^{\infty} e^{-v}\,\mathrm{d}v \\ &= \lambda^{-1/\alpha}\,\Gamma\big(1 + \tfrac{1}{\alpha}, \lambda t^{\alpha}\big) - t\, e^{-\lambda t^{\alpha}}, \end{aligned}$$

where $\Gamma(s, x) = \int_x^{\infty} v^{s-1} e^{-v}\,\mathrm{d}v$ is the upper incomplete gamma function. Dividing by $S(t) = e^{-\lambda t^{\alpha}}$:

$$m(t) = \frac{N(t)}{S(t)} = \frac{1}{S(t)} \cdot \frac{1}{\lambda^{1/\alpha}}\,\Gamma\big(1 + \tfrac{1}{\alpha}, \lambda t^{\alpha}\big) - t.$$

- $\alpha = 1$: $\Gamma(2, \lambda t) = (1 + \lambda t)e^{-\lambda t}$, so $m(t) = \frac{(1+\lambda t)}{\lambda} - t = \frac{1}{\lambda}$ (constant).
- $\alpha > 1$: the hazard $h(t) = \lambda\alpha t^{\alpha-1}$ is increasing, hence $m(t)$ decreases in t (proved in §10). Interpretation: the system *ages* — the longer it has survived, the shorter its expected remaining life.

10 mrl and Hazard Monotonicity

Motivation. Intuitively, if the risk of failure grows with age, the expected remaining life should shrink. This section makes that precise.

Theorem 2.19

Let $T \geq 0$ have survival S, density f, hazard $h = f/S$, and $m(t) = \mathrm{E}[T - t \mid T > t]$. Then

1. $m'(t) = h(t)m(t) - 1$;
2. if h is increasing then m is decreasing; if h is decreasing then m is increasing.

(1) Differential equation [Supplement]. $m(t) = \int_t^\infty S(u)\,\mathrm{d}u / S(t)$. By the quotient rule and $\frac{\mathrm{d}}{\mathrm{d}t}\int_t^\infty S(u)\,\mathrm{d}u = -S(t)$,

$$m'(t) = \frac{-S(t)\cdot S(t) - \left(\int_t^\infty S(u)\,\mathrm{d}u\right) S'(t)}{S(t)^2}.$$

Since $S'(t) = -f(t) = -h(t)S(t)$,

$$m'(t) = \frac{-S(t)^2 + \left(\int_t^\infty S(u)\,\mathrm{d}u\right) h(t)S(t)}{S(t)^2} = -1 + h(t)\frac{\int_t^\infty S(u)\,\mathrm{d}u}{S(t)} = -1 + h(t)m(t).$$

■

(2) Monotonicity [Supplement]. By (1), $m'(t) \leq 0 \iff h(t)m(t) \leq 1$ and $m'(t) \geq 0 \iff h(t)m(t) \geq 1$. Substituting $u \mapsto t + u$ and using $S(t+u)/S(t) = \exp\{-(H(t+u) - H(t))\}$,

$$m(t) = \int_0^\infty \frac{S(t+u)}{S(t)}\,\mathrm{d}u = \int_0^\infty \exp\Big(-\int_t^{t+u} h(s)\,\mathrm{d}s\Big)\,\mathrm{d}u.$$

- If h is increasing, $h(s) \geq h(t)$ on $[t, t+u]$, so $\int_t^{t+u} h(s)\,\mathrm{d}s \geq u\,h(t)$ and
$$m(t) \leq \int_0^\infty e^{-uh(t)}\,\mathrm{d}u = \frac{1}{h(t)} \;\Rightarrow\; h(t)m(t) \leq 1 \;\Rightarrow\; m'(t) \leq 0.$$
- If h is decreasing, $\int_t^{t+u} h(s)\,\mathrm{d}s \leq u\,h(t)$, so $m(t) \geq 1/h(t)$, $h(t)m(t) \geq 1$, and $m'(t) \geq 0$.

■

Example 2.20: Application to the Weibull

$h(t) = \lambda\alpha t^{\alpha-1}$.

- $\alpha < 1$: h decreasing $\Rightarrow$ m increasing.
- $\alpha = 1$: $h \equiv \lambda \Rightarrow m'(t) = \lambda m(t) - 1$; the bounded solution is $m \equiv 1/\lambda$ (so $m' = 0$).
- $\alpha > 1$: h increasing $\Rightarrow$ m decreasing.

11 Selected exercises

Motivation. Three textbook exercises (K&M Section 2.8) that use the relations of this lecture in less familiar directions: a discrete constant hazard, a population mixture whose hazard falls although every individual hazard is constant, and recovering a whole distribution from its mean residual life.

Exercise 2.1 (K&M 2.13: geometric distribution)

Let $P(X = x) = p(1-p)^{x-1}$, $x = 1, 2, \ldots$, with $0 < p < 1$.
(a) Find the survival function of X.
(b) Find the discrete hazard of X and compare it with the exponential hazard.

Solution.

(a) For an integer $k \geq 0$, the geometric series gives

$$P(X > k) = \sum_{j=k+1}^{\infty} p(1-p)^{j-1} = p(1-p)^k \sum_{i=0}^{\infty} (1-p)^i = (1-p)^k.$$

Since X is integer valued, $S(x) = P(X > \lfloor x \rfloor) = (1-p)^{\lfloor x \rfloor}$ for $x \geq 0$.

(b) By the definition of the discrete hazard, for $j \geq 1$,

$$h(j) = \frac{P(X = j)}{P(X \geq j)} = \frac{p(1-p)^{j-1}}{S(j-1)} = \frac{p(1-p)^{j-1}}{(1-p)^{j-1}} = p,$$

a constant, as for the exponential. Equivalently $P(X > j + k \mid X > j) = (1-p)^k$ does not depend on j (discrete memorylessness). At the integers $S(j) = e^{-\lambda j}$ with $\lambda = -\log(1-p)$, so the matching exponential rate is $\lambda = -\log(1-p) > p$. This is exactly the Cox–Oakes cumulative hazard $-\sum_{i \leq j} \log(1 - h(i)) = j\lambda$, whereas the K&M definition gives $\sum_{i \leq j} h(i) = jp$, and $e^{-jp} \neq S(j)$. ■

Exercise 2.2 (K&M 2.14: gamma mixture of exponentials)

Given $\Theta = \theta$, the lifetime X is exponential with hazard θ; the individual hazard Θ has the gamma density $\lambda^\beta \theta^{\beta-1} e^{-\lambda\theta}/\Gamma(\beta)$, $\theta > 0$, with $\lambda, \beta > 0$.
(a) Find the survival function of X.
(b) Find the hazard rate of X and describe its shape.

Solution.

(a) Conditioning on Θ, $S(x) = \mathrm{E}[P(X > x \mid \Theta)] = \mathrm{E}[e^{-\Theta x}]$. For $x \geq 0$,

$$\mathrm{E}[e^{-\Theta x}] = \int_0^\infty \frac{\lambda^\beta \theta^{\beta-1} e^{-(\lambda+x)\theta}}{\Gamma(\beta)} \, \mathrm{d}\theta = \frac{\lambda^\beta}{(\lambda+x)^\beta} \int_0^\infty \frac{(\lambda+x)^\beta \theta^{\beta-1} e^{-(\lambda+x)\theta}}{\Gamma(\beta)} \, \mathrm{d}\theta = \Big(\frac{\lambda}{\lambda+x}\Big)^\beta,$$

since the last integrand is the Gamma$(\lambda + x, \beta)$ density.

(b) $H(x) = -\log S(x) = \beta\{\log(\lambda + x) - \log\lambda\}$, so

$$h(x) = H'(x) = \frac{\beta}{\lambda + x},$$

which is strictly decreasing from $h(0) = \beta/\lambda = \mathrm{E}(\Theta)$ to 0. Interpretation: every individual has a constant hazard, but the frail ones (large Θ) fail first, so the survivors at time x are increasingly the robust ones. Indeed, by Bayes' rule the density of Θ given $X > x$ is proportional to $e^{-\theta x} \theta^{\beta-1} e^{-\lambda\theta}$, the Gamma$(\lambda + x, \beta)$ density, and $h(x) = \mathrm{E}(\Theta \mid X > x) = \beta/(\lambda + x)$. (The marginal law is a Pareto-type, Lomax, distribution.) ■

Exercise 2.3 (K&M 2.17: distribution from the mean residual life)

A continuous lifetime X has mean residual life $\mathrm{mrl}(x) = x + 10$ for all $x \geq 0$.
(a) Find $\mathrm{E}(X)$.

(b) Find $h(x)$.
(c) Find $S(x)$.

Solution.

(a) $\mathrm{E}(X) = \mathrm{mrl}(0) = 10$.

(b)

Step 1. Let $R(x) = \int_x^\infty S(u)\,\mathrm{d}u$. The mrl is defined only where $S(x) > 0$, so $S > 0$ on $[0, \infty)$ and $R(x) = (x + 10)S(x)$ for all $x \geq 0$.

Step 2. R is continuous, hence so is $S = R/(x+10)$; then $R'(x) = -S(x)$ exists, and $S = R/(x+10)$ is differentiable as well.

Step 3. Differentiating $R(x) = (x + 10)S(x)$ gives $-S(x) = S(x) + (x + 10)S'(x)$, so

$$h(x) = -\frac{S'(x)}{S(x)} = \frac{2}{x + 10}.$$

(Equivalently, the theorem of §10 gives $1 = m'(x) = h(x)m(x) - 1$.)

(c) $H(x) = \int_0^x \frac{2}{u+10}\,\mathrm{d}u = 2\log\frac{x+10}{10}$, so

$$S(x) = e^{-H(x)} = \Big(\frac{10}{x + 10}\Big)^2, \qquad x \geq 0.$$

Check: $\int_x^\infty 100(u+10)^{-2}\,\mathrm{d}u = 100/(x+10) = (x+10)S(x)$, as required. The increasing mrl matches the decreasing hazard, in line with §10. ■

Lecture 3. Basic Parametric Distributions and Models

1 Outline

- Basic parametric distributions for survival data: exponential, Weibull, extreme value, log-normal, log-logistic, gamma.
- Popular regression models for survival data: accelerated failure time (log-linear), multiplicative (proportional hazards), additive hazards.

2 Exponential Distribution

Definition 3.1: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$, $S(x) = e^{-\lambda x}$, $h(x) = \lambda$ (constant), $H(x) = \lambda x$.

- The conditional probability of failure at any time t, given no event before t, does not depend on t (memoryless).
- $H(x) = -\ln S(x) = \lambda x$ gives an **empirical check**: plot an estimate of $H(x)$ against x; a straight line through the origin supports the exponential model.
- Mathematically simple but too restrictive in both health and industrial applications.
- Special case of both the Weibull ($\alpha = 1$) and the gamma ($\beta = 1$).

3 Weibull Distribution

Definition 3.2: Weibull(λ, α), $\lambda, \alpha > 0$

$$h(x) = \lambda\alpha x^{\alpha-1}, \qquad H(x) = \lambda x^{\alpha}, \qquad S(x) = \exp(-\lambda x^{\alpha}).$$

- Mathematically simple, yet fairly flexible: increasing hazard ($\alpha > 1$), decreasing ($\alpha < 1$), constant ($\alpha = 1$).
- **Empirical check**: $\ln H(x) = \ln\lambda + \alpha \ln x$, so plot $\ln[\hat{H}(x)]$ against $\ln x$ and look for linearity (slope α, intercept $\ln\lambda$).

4 Weibull and the Extreme Value Distribution

Motivation. The exponential model has constant hazard, which is rarely realistic. The Weibull distribution adds a shape parameter so the hazard can increase or decrease, and on the log scale it becomes a location–scale model, the basis of accelerated failure time regression.

Definition 3.3: Standard extreme value distribution

$$f_W(w) = \exp\{w - e^w\}, \qquad S_W(w) = \exp\{-e^w\}, \qquad -\infty < w < \infty.$$

It is a special case of the Gompertz family.

Theorem 3.4: Log-linear representation of the Weibull

If $Y \equiv \ln X = \mu + \sigma W$ with W standard extreme value, then $X \sim \text{Weibull}(\lambda, \alpha)$ with

$$\alpha = \frac{1}{\sigma}, \qquad \lambda = e^{-\mu/\sigma} \quad \Big(\text{equivalently } \mu = \frac{-\ln \lambda}{\alpha},\ \sigma = \frac{1}{\alpha}\Big).$$

Proof [Supplement]. First, $S_W(w) = \int_w^\infty e^{v-e^v}\,dv$; substituting $u = e^v$ gives $\int_{e^w}^\infty e^{-u}\,du = \exp(-e^w)$. Then

$$S_X(x) = P(\ln X > \ln x) = P\Big(W > \frac{\ln x - \mu}{\sigma}\Big) = \exp\Big\{-e^{(\ln x-\mu)/\sigma}\Big\} = \exp\Big\{-e^{-\mu/\sigma}x^{1/\sigma}\Big\},$$

which is $\exp(-\lambda x^\alpha)$ with $\alpha = 1/\sigma$, $\lambda = e^{-\mu/\sigma}$. ■

Hence, if the error in a log-linear model is standard extreme value, X is Weibull.

5 Log-normal Distribution

Definition 3.5

If $Y = \ln X \sim N(\mu, \sigma^2)$ then $X \sim$ log-normal(μ, σ^2): $S(x) = 1 - \Phi\Big(\frac{\ln x - \mu}{\sigma}\Big)$.

- Close tie to the normal distribution.
- Can fit unimodal (hump-shaped) hazards: the hazard starts at 0, rises to a maximum and then decreases.
- Criticized as a lifetime distribution because the hazard *always decreases at the tail* (to 0).
- May still fit cases where large values of x are not of interest.

6 Log-logistic Distribution

Motivation. Some hazards rise and then fall, for example after surgery. The log-logistic distribution allows this and has closed-form survival and hazard functions.

Definition 3.6: Log-logistic

If $Y = \ln X = \mu + \sigma W$ with W standard logistic, LG$(0, 1)$:

$$f_W(w) = \frac{e^w}{(1+e^w)^2}, \qquad S_W(w) = \frac{1}{1+e^w},$$

then $Y \sim \text{LG}(\mu, \sigma^2)$ with $f_Y(y) = \dfrac{e^{(y-\mu)/\sigma}}{\sigma\big(1+e^{(y-\mu)/\sigma}\big)^2}$ and $X \sim$ log-logistic.

Derivation: survival function of X

$$S_X(x) = S_W\Big(\frac{\ln x - \mu}{\sigma}\Big) = \frac{1}{1+\exp\left(\frac{\ln x-\mu}{\sigma}\right)} = \frac{1}{1+e^{-\mu/\sigma}x^{1/\sigma}} = \frac{1}{1+\lambda x^{\alpha}},$$

with $\alpha = 1/\sigma$ and $\lambda = e^{-\mu/\sigma}$.

Theorem 3.7: Log-logistic hazard

$$h_X(x) = \frac{\lambda\alpha x^{\alpha-1}}{1+\lambda x^{\alpha}}.$$

It is monotone decreasing when $\alpha \le 1$. When $\alpha > 1$ it increases to a maximum at $x^* = \{(\alpha-1)/\lambda\}^{1/\alpha}$ and then decreases to 0 as $x \to \infty$.

Proof [Supplement]. $H(x) = -\ln S(x) = \ln(1+\lambda x^{\alpha})$, so $h(x) = H'(x) = \lambda\alpha x^{\alpha-1}/(1+\lambda x^{\alpha})$. Differentiating,

$$h'(x) = \lambda\alpha\,\frac{(\alpha-1)x^{\alpha-2}(1+\lambda x^{\alpha}) - x^{\alpha-1}\lambda\alpha x^{\alpha-1}}{(1+\lambda x^{\alpha})^2} = \lambda\alpha\,\frac{x^{\alpha-2}\left[(\alpha-1)-\lambda x^{\alpha}\right]}{(1+\lambda x^{\alpha})^2}.$$

If $\alpha \le 1$ the bracket is negative for all $x > 0$. If $\alpha > 1$ the bracket is positive for $\lambda x^{\alpha} < \alpha - 1$ and negative afterwards, giving the maximum at x^*; and $h(x) \sim \alpha/x \to 0$. ■

The hazard resembles the log-normal one but has a closed form.

7 Gamma Distribution

Definition 3.8: Gamma(λ, β), $\lambda, \beta > 0$

$$f(x) = \frac{\lambda^{\beta}x^{\beta-1}e^{-\lambda x}}{\Gamma(\beta)}, \qquad S(x) = \int_x^{\infty}\frac{\lambda^{\beta}s^{\beta-1}e^{-\lambda s}}{\Gamma(\beta)}\,\mathrm{d}s, \qquad h(x) = \frac{x^{\beta-1}e^{-\lambda x}}{\int_x^{\infty}s^{\beta-1}e^{-\lambda s}\,\mathrm{d}s}.$$

No closed form for S or h in general.

- $\beta = 1$: exponential(λ). $\beta \to \infty$: approximately normal. $\lambda = 1/2$, integer β: $\chi^2_{2\beta}$.
- $\beta > 1$: h is monotone increasing with $h(0) = 0$ and $h(x) \to \lambda$ as $x \to \infty$.
- $\beta < 1$: h is monotone decreasing with $h(0) = \infty$ and $h(x) \to \lambda$ as $x \to \infty$.
- Not as mathematically tractable as the others; not widely used (the Weibull is more popular).

Derivation: why the gamma hazard tends to λ monotonically

Substitute $s = x + u$ in the reciprocal hazard:

$$\frac{1}{h(x)} = \int_x^{\infty}\Big(\frac{s}{x}\Big)^{\beta-1}e^{-\lambda(s-x)}\,\mathrm{d}s = \int_0^{\infty}\Big(1+\frac{u}{x}\Big)^{\beta-1}e^{-\lambda u}\,\mathrm{d}u.$$

As $x \to \infty$, $(1+u/x)^{\beta-1} \to 1$, so $1/h(x) \to \int_0^{\infty}e^{-\lambda u}\,\mathrm{d}u = 1/\lambda$. For $\beta > 1$, $(1+u/x)^{\beta-1}$ decreases in x, so $1/h$ decreases and h increases; for $\beta < 1$ the reverse holds.

8 Regression Models for Survival Data

To deal with heterogeneity in the population, the quantities summarizing survival must be adjusted for covariates (explanatory variables). For example, in non-randomized breast cancer studies the two groups may be imbalanced in age, tumour size, number of positive nodes, or hormonal status.

8.1 Approach 1: Log-linear / accelerated failure time (AFT) model

Definition 3.9: Log-linear model

$$Y \equiv \ln X = \mu + \boldsymbol{\gamma}'\mathbf{z} + \sigma W,$$

where $W \sim F$ (a known distribution) and $\mu, \sigma, \boldsymbol{\gamma}$ are unknown.

Theorem 3.10: Log-linear ⇒ AFT

Let $X_0 = \exp(\mu + \sigma W)$ with survival S_0. Then

$$S(x \mid \mathbf{z}) = S_0(x\, e^{-\boldsymbol{\gamma}'\mathbf{z}}), \qquad h(x \mid \mathbf{z}) = \eta\, h_0(\eta x), \quad \eta = e^{-\boldsymbol{\gamma}'\mathbf{z}}.$$

Proof [Supplement].

$$\begin{aligned} P(X > x \mid \mathbf{z}) &= P(Y > \ln x \mid \mathbf{z}) = P(\mu + \sigma W > \ln x - \boldsymbol{\gamma}'\mathbf{z}) \\ &= P(\exp(\mu + \sigma W) > x \exp(-\boldsymbol{\gamma}'\mathbf{z})) \\ &= S_0(xe^{-\boldsymbol{\gamma}'\mathbf{z}}). \end{aligned}$$

For the hazard, $S(x \mid \mathbf{z}) = S_0(\eta x)$ gives $f(x \mid \mathbf{z}) = \eta f_0(\eta x)$ (chain rule), so

$$h(x \mid \mathbf{z}) = \eta f_0(\eta x)/S_0(\eta x) = \eta h_0(\eta x).$$

■

- The covariate rescales time: time is **accelerated** if $\eta = e^{-\boldsymbol{\gamma}'\mathbf{z}} > 1$ and **decelerated** if $\eta < 1$.
- The log-linear model induces the AFT model but with **opposite signs** for the regression coefficients: a positive γ lengthens survival ($\eta < 1$).

Derivation: interpretation of γ

For covariate values $\mathbf{z}_1, \mathbf{z}_2$, since $X = e^{\mu + \boldsymbol{\gamma}'\mathbf{z}} e^{\sigma W}$,

$$\frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \frac{e^{\mu + \boldsymbol{\gamma}'\mathbf{z}_1}\mathrm{E}(e^{\sigma W})}{e^{\mu + \boldsymbol{\gamma}'\mathbf{z}_2}\mathrm{E}(e^{\sigma W})} = e^{\boldsymbol{\gamma}'(\mathbf{z}_1 - \mathbf{z}_2)} \implies \ln \frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \boldsymbol{\gamma}'(\mathbf{z}_1 - \mathbf{z}_2).$$

So γ is the log of the ratio of mean survival times for a unit increase in z. (The same holds for any quantile.)

Definition 3.11: Three choices of F

1. $W \sim N(0,1)$: $\ln X \sim N(\mu + \boldsymbol{\gamma}'\mathbf{z}, \sigma^2)$, i.e. X is **log-normal**.
2. W standard extreme value, $f_W(w) = \exp(w - e^w)$: $X \sim \text{Weibull}(\lambda, \alpha)$ with $\alpha = 1/\sigma$

and $\lambda = \exp\{-(\frac{\mu}{\sigma} + \frac{\gamma'\mathbf{z}}{\sigma})\}$.

3. W standard logistic: the **log-logistic** regression model.

8.2 Approach 2: Multiplicative (proportional hazards) model

Definition 3.12: Multiplicative hazards model

$$h(x \mid \mathbf{z}) = h_0(x)\, C(\boldsymbol{\beta}'\mathbf{z}),$$

where the baseline hazard $h_0(x) \geq 0$ may be parametric or left unspecified. With $C(\boldsymbol{\beta}'\mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}}$ this is the **Cox proportional hazards model**.

- β is the **log hazard ratio**, constant over time: $h(x \mid \mathbf{z}_1)/h(x \mid \mathbf{z}_2) = e^{\boldsymbol{\beta}'(\mathbf{z}_1 - \mathbf{z}_2)}$.
- Integrating, $H(x \mid \mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)$ and $S(x \mid \mathbf{z}) = \exp\{-e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)\} = \{S_0(x)\}^{e^{\boldsymbol{\beta}'\mathbf{z}}}$.
- **Graphical check** of PH: $\ln[-\ln S(x \mid \mathbf{z})] = \boldsymbol{\beta}'\mathbf{z} + \ln[-\ln S_0(x)]$, so log-cumulative-hazard curves for different $\mathbf{z}$ should be *parallel* (vertical shift $\boldsymbol{\beta}'\mathbf{z}$).

Example 3.13: Weibull baseline under PH

If $h_0(x) = \alpha\lambda x^{\alpha-1}$ then $h(x \mid \mathbf{z}) = \alpha\lambda x^{\alpha-1}\exp(\boldsymbol{\beta}'\mathbf{z})$ and

$$S(x \mid \mathbf{z}) = \exp\big[-\lambda \exp(\boldsymbol{\beta}'\mathbf{z}) x^{\alpha}\big], \quad \text{i.e.} \quad X \mid \mathbf{z} \sim \text{Weibull}(\lambda e^{\boldsymbol{\beta}'\mathbf{z}}, \alpha).$$

Theorem 3.14: Weibull is closed under both AFT and PH

The Weibull is the only continuous distribution that is closed under both the AFT and PH models.

Derivation: matching the two parametrizations

Under AFT with Weibull baseline, $h(x \mid \mathbf{z}) = \eta h_0(\eta x) = \eta\,\alpha\lambda(\eta x)^{\alpha-1} = \eta^{\alpha}\cdot\alpha\lambda x^{\alpha-1}$, a PH model with hazard ratio $\eta^{\alpha} = e^{-\alpha\boldsymbol{\gamma}'\mathbf{z}}$. Hence

$$\boldsymbol{\beta} = -\alpha\boldsymbol{\gamma} = -\boldsymbol{\gamma}/\sigma.$$

8.3 Approach 3: Additive hazards model

Definition 3.15: Additive hazards model

$$h(x \mid \mathbf{z}) = h_0(x) + \sum_{j=1}^{p} z_j(x)\beta_j(x).$$

- Allows covariate effects to vary over time.
- The regression functions may be positive or negative, so they must be constrained to keep $h(x \mid \mathbf{z}) \geq 0$.

9 Selected exercises

Motivation. Three textbook exercises (K&M Section 2.8): a family whose hazard can be bathtub shaped, the property that characterizes log-logistic regression, and a computation with a guarantee time.

Exercise 3.1 (K&M 2.4: exponential power distribution)

Let $S(x) = \exp\{1 - \exp[(\lambda x)^\alpha]\}$, $x \geq 0$, with $\lambda, \alpha > 0$.
(a) If $\alpha = 0.5$, show that the hazard is bathtub shaped and find where it changes from decreasing to increasing.
(b) If $\alpha = 2$, show that the hazard is monotone increasing.

Solution.

Step 1 (the hazard). $H(x) = -\log S(x) = \exp[(\lambda x)^\alpha] - 1$, which is 0 at $x = 0$, so

$$h(x) = H'(x) = \alpha\lambda^\alpha x^{\alpha-1} \exp[(\lambda x)^\alpha], \qquad x > 0.$$

Step 2 (its log-derivative). For $x > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x} \log h(x) = \frac{\alpha - 1}{x} + \alpha\lambda^\alpha x^{\alpha-1} = \frac{(\alpha - 1) + \alpha(\lambda x)^\alpha}{x}.$$

Since $h > 0$, h increases exactly where the numerator $N(x) = (\alpha - 1) + \alpha(\lambda x)^\alpha$ is positive.

(a) For $\alpha = 0.5$, $N(x) = \frac{1}{2}\{(\lambda x)^{1/2} - 1\}$, negative for $x < 1/\lambda$ and positive for $x > 1/\lambda$. So h decreases on $(0, 1/\lambda]$ and increases on $[1/\lambda, \infty)$; moreover $h(x) \to \infty$ as $x \downarrow 0$ (factor $x^{-1/2}$) and as $x \to \infty$ (factor $e^{(\lambda x)^{1/2}}$). This is a bathtub shape with minimum at $x^* = 1/\lambda$. In general, for $\alpha < 1$ the turning point is $x^* = \lambda^{-1}\{(1 - \alpha)/\alpha\}^{1/\alpha}$.

(b) For $\alpha = 2$, $N(x) = 1 + 2(\lambda x)^2 > 0$ for all x, so h is strictly increasing (more generally whenever $\alpha \geq 1$). ■

Exercise 3.2 (K&M 2.16: log-logistic regression is a proportional odds model)

Given a covariate Z, let $Y = \ln X = \mu + \gamma Z + \sigma W$ with $\sigma > 0$ and W standard logistic, $f_W(w) = e^w/(1 + e^w)^2$.
(a) Find $S(x \mid Z)$.
(b) Find the odds of death before x, $[1 - S(x \mid Z)]/S(x \mid Z)$.
(c) Show that for two covariate values Z_1, Z_2 the ratio of these odds does not depend on x.

Solution.

(a) With $S_W(w) = 1/(1 + e^w)$ (Definition of the log-logistic in this lecture),

$$S(x \mid Z) = P\Big(W > \frac{\ln x - \mu - \gamma Z}{\sigma}\Big) = \frac{1}{1 + \exp\{(\ln x - \mu - \gamma Z)/\sigma\}} = \frac{1}{1 + \lambda_Z x^\alpha},$$

where $\alpha = 1/\sigma$ and $\lambda_Z = \exp\{-(\mu + \gamma Z)/\sigma\}$.

(b) $1 - S(x \mid Z) = \lambda_Z x^\alpha/(1 + \lambda_Z x^\alpha)$, so the odds are $\lambda_Z x^\alpha$.

(c) The ratio is

$$\frac{\lambda_{Z_1} x^\alpha}{\lambda_{Z_2} x^\alpha} = \exp\Big\{-\frac{\gamma(Z_1 - Z_2)}{\sigma}\Big\},$$

free of x. Thus the log-logistic AFT model is also a *proportional odds* model, with log odds ratio $-\gamma(Z_1 - Z_2)/\sigma$, in the same way that the Weibull AFT model is a PH model with $\boldsymbol{\beta} = -\boldsymbol{\gamma}/\sigma$. ■

Exercise 3.3 (K&M 2.11: Weibull with a guarantee time)

For $\phi \geq 0$ let $S(x) = \exp[-\lambda(x-\phi)^\alpha]$ for $x > \phi$ and $S(x) = 1$ for $x \leq \phi$.
(a) Find the hazard and density.
(b) With $\alpha = 1$, $\lambda = 0.0075$, $\phi = 100$, find the mean and the median lifetime.

Solution.

(a) For $x \leq \phi$, $S \equiv 1$ gives $f(x) = h(x) = 0$. For $x > \phi$, $H(x) = \lambda(x-\phi)^\alpha$, so

$$h(x) = \alpha\lambda(x-\phi)^{\alpha-1}, \qquad f(x) = h(x)S(x) = \alpha\lambda(x-\phi)^{\alpha-1}\exp[-\lambda(x-\phi)^\alpha].$$

That is, $X - \phi \sim \text{Weibull}(\lambda, \alpha)$: no failure can occur before the guarantee time ϕ.

(b) With $\alpha = 1$, $X - 100 \sim \text{Exp}(0.0075)$. Hence

$$\text{E}(X) = 100 + \frac{1}{0.0075} = 233.3, \qquad x_{0.5} = 100 + \frac{\ln 2}{0.0075} = 100 + 92.4 = 192.4.$$

■

Lecture 4. Formal Introduction to Censoring and Truncation

1 Right Censoring

Definition 4.1: Right-censored data

An observation is right censored if it is not known exactly, only that it is greater than some value. Let X = time to the event of interest and C = censoring time. We observe

$$(T, \delta), \qquad T = \min(X, C), \qquad \delta = I(X < C) \text{ (or } I(X \leq C)),$$

and the data are n realizations $\{(T_i, \delta_i),\ i = 1, \ldots, n\}$.

There are many types of right censoring; some of them provide information on C even when $X < C$.

1.1 Type I censoring

The event is observed only if it occurs before some *pre-specified* time.

- **Fixed Type I censoring**: $C_i = \tau$ for all i. Typical of controlled experiments, e.g. a mice study of time to death after a fixed period following intense carcinogen exposure: every animal still alive at the end of the study is censored at the same time.
- **Progressive Type I censoring**: C is fixed in advance but may take different values for each i, with only a finite number of distinct values. *Example*: X = time to a certain % weight loss, weights measured every 12 hours; mice in cages 1–2 are sacrificed at the end of week 1 and those in cages 3–4 by the end of week 2. (A second example uses two sacrifice times, 42 and 104 weeks.)
- **Generalized Type I censoring**: C may differ for every individual but is known in advance. Common in human studies with *staggered entry* and complete follow-up to the end of the study, e.g. a Phase I chemotherapy study with X = time to toxicity: C_i = (end of study) − (entry time of i) is computable at enrolment, assuming a fixed study length and no dropout.

1.2 Type II censoring

- **Simple Type II**: sample size n; the study continues until the first r failures. The number of failures r and censored observations $n - r$ are *fixed*; the censoring time $C = X_{(r)}$ is *random*.
- **Progressive Type II**: after the first r_1 failures, $n_1 - r_1$ of the remaining $n - r_1$ unfailed items are removed (sacrificed); after the next r_2 failures, $n_2 - r_2$ of the unfailed items are removed; and so on until a pre-decided number of repetitions is completed.

Remark 4.2: Type I vs. Type II

Type I: C is fixed (predetermined before the study starts). Type II: C is random, namely the time at which a predetermined number of events has occurred. This distinction matters when constructing the likelihood.

1.3 Random censoring

Typical of human studies: $C \sim G(t)$, not fixed in advance and possibly unknown.

- Scenario 1: censoring at the end of study ("administrative loss to follow-up"), fixed in advance, similar to Type I.
- Scenario 2: censoring before the end of study ("drop-out"), random, due to other factors.
- In practice censoring is a mixture of the two.

2 Left, Double and Interval Censoring

Definition 4.3: Left censoring

An observation is left censored if it is not known exactly, only that it is less than some value C_l. The exact lifetime is known iff $X \geq C_l$. Data: (T, δ) with $T = \max(X, C_l)$, $\delta = I(X \geq C_l)$.

Definition 4.4: Double censoring

With left censoring time C_l and right censoring time C_r $(C_l < C_r)$:

$$T = \max\{\min(X, C_r),\, C_l\}, \qquad \delta = \begin{cases} 1, & C_l \leq X \leq C_r \text{ (exact)} \\ -1, & X < C_l \text{ (left censored)} \\ 0, & X > C_r \text{ (right censored)} \end{cases}$$

Remark 4.5: Double vs. interval censoring

Interval censoring: each observation is only known to lie in an interval, $L < X \leq R$; no exact time is observed. **Double censoring**: each observation is one of three possibilities — fully observed, right censored, or left censored.

3 Truncation

Definition 4.6: Truncation

Only individuals whose event time lies within an observational window (Y_L, Y_R) are observed.

- $Y_R = \infty$: **left truncation**; we observe only those with $Y_L < X$.
- $Y_L = 0$: **right truncation**; we observe only those with $X \leq Y_R$.

Inference for truncated data is restricted to *conditional* estimation.

Example 4.7: Breast cancer onset (classification exercise)

To estimate the distribution of age at breast cancer onset in post-menopausal women, 8 healthy 50-year-old women had yearly mammograms for 10 years; no tumours were detected between scheduled exams. Onset ages:

$$(55, 56],\ (58, 59],\ (52, 53],\ (59, 60],\ > 60,\ > 60,\ > 60,\ > 60.$$

Answer. The four detected cases are **interval censored** (onset between two yearly exams). The four undetected cases are **right censored** at 60. Moreover, only women who were cancer-free at age 50 were enrolled, so the ages are **left truncated** at 50 (inference is conditional on $X > 50$).

4 Likelihood Construction

4.1 Contributions of each observation type

Let X be continuous with pdf f, cdf F and $S = 1 - F$; C_r = right censoring point, C_l = left censoring point, $(L, R]$ = censoring interval, Y = truncation point.

Observation	Likelihood contribution
Exact lifetime	$f(x)$
Right censored	$S(C_r) = P(X > C_r)$
Left censored	$1 - S(C_l) = P(X < C_l)$
Interval censored	$S(L) - S(R) = P(L < X \leq R)$
Left truncated	$f(x)/S(Y_L)$
Right truncated	$f(x)/[1 - S(Y_R)]$
Interval truncated	$f(x)/[S(Y_L) - S(Y_R)]$

The truncated contributions are conditional densities: e.g. for left truncation, $f(x \mid X > Y_L) = f(x)/P(X > Y_L) = f(x)/S(Y_L)$.

Theorem 4.8: General censored-data likelihood

$$L \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_{r,i}) \prod_{i \in LC} F(C_{l,i}) \prod_{i \in IC} \{S(L_i) - S(R_i)\},$$

where D = exact (death) times, $RC/LC/IC$ = right-/left-/interval-censored observations.

Truncation. For left-truncated data (possibly also right censored), replace $f(x_i)$ by $f(x_i)/S(Y_i)$ and $S(C_{r,i})$ by $S(C_{r,i})/S(Y_i)$. For right-truncated data (only events observed), replace $f(x_i)$ by $f(x_i)/[1 - S(Y_i)]$.

4.2 Type I right censoring

Theorem 4.9

Observing (T_i, δ_i) with $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and fixed C_i,

$$L = \prod_{i=1}^{n} \Pr(t_i, \delta_i) = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i} = \prod_{i=1}^{n} h(t_i)^{\delta_i} e^{-H(t_i)}.$$

Proof [Supplement]. Since C_i is fixed, $\delta_i = 0$ means $T_i = C_i$ and $X_i > C_i$, with probability $S(C_i) = S(t_i)$; $\delta_i = 1$ means $T_i = X_i \leq C_i$ with density $f(t_i)$. For the last form, $f = hS$ and $S = e^{-H}$: $f^{\delta} S^{1-\delta} = (hS)^{\delta} S^{1-\delta} = h^{\delta} S = h^{\delta} e^{-H}$. ■

Example 4.10: Exponential with right censoring

$X \sim \text{Exp}(\lambda)$, $T = \min(X, C)$, $\delta = I(X < C)$:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

With $h(t) = \lambda$ and $H(t) = \lambda t$,

$$L(\lambda) = \prod_{i=1}^{5} \lambda^{\delta_i} e^{-\lambda t_i} = \lambda^{\sum \delta_i} e^{-\lambda \sum t_i} = \lambda^3 e^{-3.25\lambda}.$$

Maximizing $\ell(\lambda) = 3\ln\lambda - 3.25\lambda$: $\ell'(\lambda) = 3/\lambda - 3.25 = 0 \Rightarrow \hat{\lambda} = 3/3.25 \approx 0.923$. In general $\hat{\lambda} = \sum \delta_i / \sum t_i$ = (number of events)/(total time at risk).

4.3 Type II censoring

Theorem 4.11

With n i.i.d. $X_i \sim f$, observing the r smallest order statistics $X_{(1)} < \cdots < X_{(r)}$ and $n - r$ values censored at $X_{(r)}$,

$$L = \frac{n!}{(n-r)!} \prod_{i=1}^{r} f(x_{(i)}) \left\{S(x_{(r)})\right\}^{n-r}.$$

Derivation: where the constant comes from

The joint density of the first r order statistics: choose which r of the n units fail (and in which order) — $n!/(n-r)!$ ordered choices — each failed unit contributes $f(x_{(i)})$, and the remaining $n - r$ units must all survive beyond $x_{(r)}$, contributing $S(x_{(r)})^{n-r}$. The constant does not involve the parameters, so the kernel is the same as in Theorem 4.8.

Progressive Type II (two stages): first r_1 ordered failures $X_{(1)}, \ldots, X_{(r_1)}$, then $n_1 - r_1$ items removed; the next r_2 failures $X^*_{(1)}, \ldots, X^*_{(r_2)}$, then the remaining $n - n_1 - r_2$ removed:

$$L \propto \prod_{i=1}^{r_1} f(x_{(i)}) \left\{S(x_{(r_1)})\right\}^{n_1 - r_1} \prod_{i=1}^{r_2} f(x^*_{(i)}) \left\{S(x^*_{(r_2)})\right\}^{n - n_1 - r_2} \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_i).$$

4.4 Random censoring

$X_i \sim f, S$ and $C_i \sim g, G$ are both random but **independent**; we observe $T_i = \min(X_i, C_i)$ and $\delta_i = I(X_i \le C_i)$.

Theorem 4.12

$$L = \prod_{i=1}^{n} \left\{f(t_i)G(t_i)\right\}^{\delta_i} \left\{g(t_i)S(t_i)\right\}^{1-\delta_i} = \underbrace{\prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}}_{L_X} \underbrace{\prod_{i=1}^{n} g(t_i)^{1-\delta_i} G(t_i)^{\delta_i}}_{L_C}.$$

Proof [Supplement]. For $\delta = 1$: $P(T \in \mathrm{d}t, \delta = 1) = P(X \in \mathrm{d}t,\ C \ge t) = f(t)G(t)\,\mathrm{d}t$ by independence. For $\delta = 0$: $P(T \in \mathrm{d}t, \delta = 0) = P(C \in \mathrm{d}t,\ X > t) = g(t)S(t)\,\mathrm{d}t$. Multiply over i and regroup. ■

If $f = f(\cdot;\theta)$ and the distribution of C does not depend on θ (**non-informative censoring**), L_C is constant in θ and

$$L \propto L_X = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i},$$

the same form as under Type I censoring.

5 Remarks

- These likelihoods are used primarily for fitting parametric models (K&M Ch. 12).
- They also underlie the *partial likelihood* of the semiparametric Cox model (K&M Ch. 8–9).
- In most applications they are not used explicitly, but the rationale — how each data component contributes, as in Theorem 4.8 — is essential.

6 Selected exercises

Motivation. Three textbook exercises (K&M Section 3.7): writing a likelihood when truncation and two kinds of censoring occur together, the exact sampling distribution of the exponential MLE under random censoring, and the simplest counting-process martingale.

Exercise 4.1 (K&M 3.2: periodic exams with delayed entry)

Disease-free women were enrolled from January 1, 1970, and followed for 30 years for the age X at onset of breast cancer, with clinical exams every 3 years after enrollment; onset is detected only at an exam. Classify the censoring and truncation, and write the likelihood contribution, for each woman:
(a) enrolled at age 30, never developed breast cancer during the study;
(b) enrolled at age 40, diagnosed at the fifth exam after enrollment;
(c) enrolled at age 50, died of an unrelated cause at age 61 without a diagnosis;
(d) enrolled at age 42, moved away at age 55 without a diagnosis.
(e) Write the likelihood for these four women.

Solution. Every woman is enrolled only if disease-free at entry, so each observation is *left truncated* at her entry age v: her contribution is the probability of what is observed divided by $S(v)$ (Section on truncation above). Onset is known only up to the exam schedule $v+3, v+6, \ldots$

(a) Exams at ages $33, \ldots, 60$, all negative: right censored at 60, left truncated at 30. Contribution $S(60)/S(30)$.

(b) The fourth exam (age 52) was negative and the fifth (age 55) positive: interval censored in $(52, 55]$, left truncated at 40. Contribution $\{S(52) - S(55)\}/S(40)$.

(c) Exams at $53, 56, 59$ were negative; death at 61 is a competing event that ends follow-up. Onset could have occurred unseen between 59 and 61, so she is right censored at the last negative exam, 59, and left truncated at 50. Contribution $S(59)/S(50)$. (If an onset before death would surely have been noticed, one may censor at 61 instead, giving $S(61)/S(50)$.)

(d) Exams at $45, 48, 51, 54$ negative, then lost to follow-up: right censored at 54, left truncated at 42. Contribution $S(54)/S(42)$.

(e) Assuming independent censoring and truncation,

$$L = \frac{S(60)}{S(30)} \cdot \frac{S(52) - S(55)}{S(40)} \cdot \frac{S(59)}{S(50)} \cdot \frac{S(54)}{S(42)}.$$

■

Exercise 4.2 (K&M 3.8: exponential lifetimes, exponential censoring)

Let $X \sim \text{Exp}(\lambda)$ and $C \sim \text{Exp}(\theta)$ be independent, $T = \min(X, C)$, $\delta = I(X \le C)$.
(a) Find $P(\delta = 1)$.
(b) Find the distribution of T.
(c) Show that δ and T are independent.
(d) For a random sample (T_i, δ_i), $i = 1, \ldots, n$, show that the MLE of λ is $\hat{\lambda} = \sum_i \delta_i / \sum_i T_i$, and use (a)–(c) to find its mean ($n \ge 2$) and variance ($n \ge 3$).

Solution.

(a)–(c) For $t \ge 0$, by independence,

$$P(\delta = 1,\, T > t) = P(t < X \le C) = \int_t^\infty \lambda e^{-\lambda s} e^{-\theta s}\, \mathrm{d}s = \frac{\lambda}{\lambda + \theta} e^{-(\lambda+\theta)t}.$$

With $t = 0$, $P(\delta = 1) = \lambda/(\lambda + \theta)$. Also $P(T > t) = P(X > t)P(C > t) = e^{-(\lambda+\theta)t}$, so $T \sim \text{Exp}(\lambda + \theta)$. Hence $P(\delta = 1, T > t) = P(\delta = 1)P(T > t)$ for all t; since δ is binary, this gives $P(\delta = 0, T > t) = P(\delta = 0)P(T > t)$ too, so $\delta \perp\!\!\!\perp T$.

(d)

Step 1 (MLE). By the random-censoring theorem above, $L = \lambda^D e^{-\lambda G} \cdot \theta^{n-D} e^{-\theta G}$ with $D = \sum_i \delta_i$ and $G = \sum_i T_i$. The λ-part has $\partial_\lambda \log L = D/\lambda - G$, positive for $\lambda < D/G$ and negative after, so $\hat{\lambda} = D/G$ (when $D = 0$ the supremum is at the boundary $\hat{\lambda} = 0$).

Step 2 (distributions). The pairs are i.i.d. and $\delta_i \perp\!\!\!\perp T_i$ by (c), so $\delta_1, \ldots, \delta_n, T_1, \ldots, T_n$ are mutually independent. Hence $D \sim \text{Bin}(n, p)$ with $p = \lambda/r$, $r = \lambda + \theta$, $G \sim \text{Gamma}(r, n)$ (shape n, rate r), and $D \perp\!\!\!\perp G$.

Step 3 (inverse moments of G). For $k < n$,

$$\mathrm{E}(G^{-k}) = \int_0^\infty g^{-k} \frac{r^n g^{n-1} e^{-rg}}{\Gamma(n)}\, \mathrm{d}g = \frac{r^k\, \Gamma(n-k)}{\Gamma(n)},$$

so $\mathrm{E}(G^{-1}) = r/(n-1)$ and $\mathrm{E}(G^{-2}) = r^2/\{(n-1)(n-2)\}$.

Step 4 (mean). By independence, $\mathrm{E}\hat{\lambda} = \mathrm{E}(D)\mathrm{E}(G^{-1}) = np \cdot r/(n-1) = n\lambda/(n-1)$, a slight upward bias.

Step 5 (variance). $\mathrm{E}(D^2) = np(1-p) + n^2p^2$, and $r^2p(1-p) = \lambda\theta$, $r^2p^2 = \lambda^2$. So

$$\mathrm{E}\hat{\lambda}^2 = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)},$$

$$\mathrm{Var}\hat{\lambda} = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)} - \frac{n^2\lambda^2}{(n-1)^2} = \frac{n\lambda\theta}{(n-1)(n-2)} + \frac{n^2\lambda^2}{(n-1)^2(n-2)}.$$

As $n \to \infty$, $n\mathrm{Var}\hat{\lambda} \to \lambda\theta + \lambda^2 = \lambda^2/p$, the inverse of the Fisher information per observation, $\mathrm{E}(\delta)/\lambda^2 = p/\lambda^2$. ■

Exercise 4.3 (K&M 3.9: the Poisson process martingale)

Let $N(t)$ be a Poisson process with rate λ: $N(0) = 0$; $N(t) - N(s) \sim \text{Poisson}(\lambda(t-s))$ for $0 \le s < t$; increments over disjoint intervals are independent. Let $\mathcal{F}_s = \sigma\{N(u) : u \le s\}$ and $M(t) = N(t) - \lambda t$.
(a) Show that $\mathrm{E}|M(t)| < \infty$.

(b) Show that $\mathrm{E}[M(t) \mid \mathcal{F}_s] = M(s)$ for $s < t$, so M is a martingale and λt is the compensator of $N(t)$.

Solution.

(a) $\mathrm{E}|M(t)| \le \mathrm{E}N(t) + \lambda t = 2\lambda t < \infty$.

(b)

Step 1 ($N(t) - N(s) \perp\!\!\!\perp \mathcal{F}_s$). For $0 \le u_1 < \cdots < u_k \le s$, the vector $(N(u_1), \ldots, N(u_k))$ is a function of the increments over $(0, u_1], (u_1, u_2], \ldots, (u_{k-1}, u_k]$, which are independent of the increment over the disjoint interval $(s, t]$. So $N(t) - N(s)$ is independent of every event in the π-system of finite-dimensional cylinder sets $\{(N(u_1), \ldots, N(u_k)) \in B\}$, which generates $\mathcal{F}_s$. By the π–λ theorem it is independent of $\mathcal{F}_s$.

Step 2. Write $M(t) = \{M(t) - M(s)\} + M(s)$. Since $M(s)$ is $\mathcal{F}_s$-measurable and integrable,

$$\mathrm{E}[M(t) \mid \mathcal{F}_s] = \mathrm{E}[N(t) - N(s)] - \lambda(t - s) + M(s) = \lambda(t - s) - \lambda(t - s) + M(s) = M(s),$$

using Step 1 to drop the conditioning.

Step 3. M is adapted, integrable by (a), and has the martingale property by Step 2. The process λt is deterministic (hence predictable), continuous and increasing with $\lambda \cdot 0 = 0$, and $N(t) - \lambda t$ is a martingale; this is the definition of the compensator of N. It is the integrated hazard $\int_0^t \lambda \, \mathrm{d}u$, the prototype of $\int_0^t Y(u)h(u) \, \mathrm{d}u$ behind the Nelson–Aalen estimator. ■

Lecture 5. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 1)

1 Setting

X is a continuous failure time with survival function $S(t)$ and cumulative hazard $H(t)$. X is subject to **non-informative right censoring**, so we observe

$$\{(T_i, \delta_i),\ i = 1, \ldots, n\}, \qquad T_i = \min(X_i, C_i), \quad \delta_i = I(X_i \le C_i).$$

Goal: estimate $S(t)$ and $H(t)$ without assuming a parametric family. We build up the estimator in four steps: discrete/continuous $\times$ without/with censoring.

2 Building Up the Estimator

2.1 Discrete X without censoring

Let X have M support points $0 < \tilde{x}_1 < \cdots < \tilde{x}_M < \infty$. With $\tilde{x}_0 = 0$, $S(\tilde{x}_0) = 1$, the product-limit representation (Lecture 2) is

$$\begin{aligned} S(x) = P(X > x) &= \prod_{\tilde{x}_j \le x} \frac{S(\tilde{x}_j)}{S(\tilde{x}_{j-1})} = \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X > \tilde{x}_{j-1}) \\ &= \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X \ge \tilde{x}_j) = \prod_{\tilde{x}_j \le x} [1 - h(\tilde{x}_j)], \qquad h(\tilde{x}_j) = P(X = \tilde{x}_j \mid X \ge \tilde{x}_j). \end{aligned}$$

(The third equality holds because there is no mass strictly between $\tilde{x}_{j-1}$ and $\tilde{x}_j$.) With i.i.d. $X_1, \ldots, X_n$, estimate

$$\hat{h}(\tilde{x}_j) = \frac{\#\{X_i = \tilde{x}_j\}}{\#\{X_i \ge \tilde{x}_j\}} = \frac{\#\text{ deaths at } \tilde{x}_j}{\#\text{ deaths at } \tilde{x}_j, \tilde{x}_{j+1}, \ldots, \tilde{x}_M}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} [1 - \hat{h}(\tilde{x}_j)].$$

$\hat{h}$ and $\hat{S}$ are the MLEs of h and S.

2.2 Discrete X with right censoring

With $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and $X_i \perp\!\!\!\perp C_i$:

$$\begin{aligned} P(T_i = \tilde{x}_j, \delta_i = 1 \mid T_i \ge \tilde{x}_j) &= \frac{P(T_i = \tilde{x}_j, \delta_i = 1)}{P(T_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j,\ C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j,\ C_i \ge \tilde{x}_j)} \\ &= \frac{P(X_i = \tilde{x}_j)P(C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)P(C_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)} = h(\tilde{x}_j). \end{aligned}$$

So the hazard is identifiable from the *observed* data, which suggests

$$\hat{h}(\tilde{x}_j) = \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} \Big[1 - \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}\Big].$$

Derivation: $\hat{h}$ is the MLE under censoring

Under independent censoring, the likelihood in the hazard parameters factorizes over support points. With d_j deaths among Y_j at risk at $\tilde{x}_j$, each subject at risk either dies at $\tilde{x}_j$ (prob. h_j) or survives past it (prob. $1-h_j$); censored subjects contribute survival up to their censoring time. Thus

$$L(h_1,\dots,h_M) \propto \prod_j h_j^{d_j}(1-h_j)^{Y_j-d_j}, \qquad \frac{\partial \log L}{\partial h_j} = \frac{d_j}{h_j} - \frac{Y_j-d_j}{1-h_j} = 0 \;\Rightarrow\; \hat{h}_j = \frac{d_j}{Y_j}.$$

By invariance, $\hat{S}(x) = \prod_{\tilde{x}_j \le x}(1-\hat{h}_j)$ is the MLE of S.

2.3 Continuous X

Without censoring, the survival function is smooth and can be estimated by the complement of the empirical distribution function:

$$\hat{S}(t) = 1 - \mathbb{F}_n(t) = 1 - \frac{1}{n}\sum_{i=1}^{n} I(X_i \le t).$$

With right censoring we must modify this. Two questions guide the construction: *Should the estimator jump at censoring times?* (No — no event is observed there.) *What is the contribution of a censored observation?* (It counts in the risk set Y_j for every death time up to its censoring time, then leaves.)

3 The Kaplan–Meier (Product-Limit) Estimator

Motivation. With censored data the empirical survival function cannot be computed, since we do not know when censored subjects fail. Kaplan–Meier writes survival as a product of conditional survival probabilities over the observed failure times, each estimated from those still at risk.

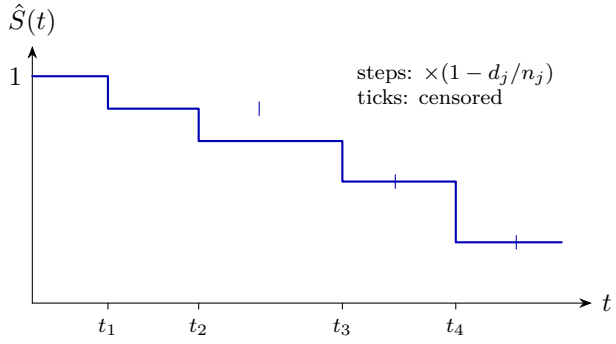


Figure. *The Kaplan–Meier estimate is a step function that drops only at observed failure times t_j, by the factor $1-d_j/n_j$. Censored observations (ticks) cause no drop but leave the risk set, which makes later drops larger.*

Definition 5.1: Notation

- $t_1 < t_2 < \cdots < t_D$: the D distinct death times;
- $d_j = \sum_{i=1}^n I(T_i = t_j, \delta_i = 1)$: number of deaths at t_j;
- $Y_j = \sum_{i=1}^n I(T_i \ge t_j)$: number at risk (alive and under observation) just before t_j.

Theorem 5.2: Kaplan–Meier estimator

$$\hat{S}(t) = \begin{cases} 1, & t < t_1, \\ \prod_{t_j \le t} \left(1 - \frac{d_j}{Y_j}\right), & t_1 \le t. \end{cases}$$

Remark 5.3: Ties between deaths and censorings

If a death and a censoring occur at the same recorded time, the convention is that the censoring happens just *after* the death, so the censored subject is still counted in Y_j (Y_j uses $T_i \ge t_j$).

Theorem 5.4: Greenwood's variance estimator

$$\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}.$$

Derivation: delta-method derivation (Cox & Oakes, 1984)

Let $\hat{h}_j = d_j/Y_j$. Treating $d_j \mid Y_j \sim \mathrm{Bin}(Y_j, h_j)$ and the $\hat{h}_j$ as approximately uncorrelated across j,

$$\mathrm{Var}(\hat{h}_j) \approx \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j}.$$

Since $\ln \hat{S}(t) = \sum_{t_j \le t} \ln(1 - \hat{h}_j)$ and, by the delta method with $g(u) = \ln(1-u)$, $g'(u) = -1/(1-u)$:

$$\begin{aligned} \mathrm{Var}\{\ln \hat{S}(t)\} &\approx \sum_{t_j \le t} \mathrm{Var}\{\ln(1-\hat{h}_j)\} \approx \sum_{t_j \le t} \left(\frac{1}{1-\hat{h}_j}\right)^2 \mathrm{Var}(\hat{h}_j) \\ &\approx \sum_{t_j \le t} \left(\frac{1}{1-\hat{h}_j}\right)^2 \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j \le t} \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j \le t} \frac{d_j/Y_j}{Y_j(Y_j-d_j)/Y_j} \\ &= \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j-d_j)}. \end{aligned}$$

A second delta-method step with $\hat{S} = \exp(\ln \hat{S})$ gives $\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \,\mathrm{Var}\{\ln \hat{S}(t)\}$.

4 The Nelson–Aalen Estimator

Motivation. Estimating the cumulative hazard by summing the observed failure rates d_i/n_i gives the Nelson–Aalen estimator, which has better small-sample properties and is the natural input for graphical model checks.

Theorem 5.5: Nelson–Aalen estimator and Aalen's variance

$$\tilde{H}(t) = \begin{cases} 0, & t < t_1, \\ \sum_{t_j \le t} \frac{d_j}{Y_j}, & t_1 \le t, \end{cases} \qquad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Remark 5.6: Intuition

$H(t) = \int_0^t h(s)\,\mathrm{d}s$ accumulates the hazard; at each death time the hazard increment is estimated by $\hat{h}_j = d_j/Y_j$. Aalen's variance treats d_j as approximately Poisson with mean $Y_j h_j$, so $\mathrm{Var}(d_j/Y_j) \approx Y_j h_j / Y_j^2 \approx d_j/Y_j^2$.

4.1 Alternative estimators

- An alternative estimator of H: $\hat{H}(t) = -\log \hat{S}(t)$.
- An alternative estimator of S: $\tilde{S}(t) = \exp\{-\tilde{H}(t)\}$ (the Breslow / Fleming–Harrington estimator).
- Since $1 - x \le e^{-x}$, each KM factor $1 - d_j/Y_j \le e^{-d_j/Y_j}$, so $\hat{S}(t) \le \tilde{S}(t)$ and equivalently $\tilde{H}(t) \le \hat{H}(t)$.
- $\hat{S}(t)$ is called the "product-limit" estimator: as $n \uparrow \infty$, $D \uparrow \infty$ and the number of terms in the product $\to \infty$ for continuous X. For finite n, $D < \infty$; in the limit the product runs over the whole support of X (a *product integral*).

5 Worked Example (Small Sample)

Example 5.7: $n = 5$

Continuous X subject to right censoring, $T = \min(X, C)$, $\delta = I(X \le C)$:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

Ordered: 0.25^+, 0.5, 0.75, 0.75^+, 1. At $t = 0.75$ the censored subject is still at risk (tie convention).

t_j	d_j	Y_j	$1 - d_j/Y_j$	$\hat{S}(t_j)$	d_j/Y_j	$\tilde{H}(t_j)$	$\tilde{S}(t_j) = e^{-\tilde{H}}$
0.5	1	4	3/4	0.750	0.250	0.250	0.779
0.75	1	3	2/3	0.500	0.333	0.583	0.558
1	1	1	0	0.000	1.000	1.583	0.205

Greenwood: $\hat{V}[\hat{S}(0.5)] = 0.75^2 \cdot \frac{1}{4\cdot 3} = 0.0469$, $\hat{V}[\hat{S}(0.75)] = 0.5^2(\frac{1}{12} + \frac{1}{3\cdot 2}) = 0.0625$; at $t = 1$, $Y_j - d_j = 0$ and Greenwood is undefined (the KM drops to 0).

6 Remarks

- $\hat{S}$, $\tilde{H}$, $\hat{H}$ are right-continuous step functions even though S and H may be continuous.
- $\hat{S}$ and $\tilde{S}$ are consistent for $S(t)$ for all $t \le \tau$ such that $P(T \ge \tau) > 0$, and are asymptotically equivalent.
- $\hat{S}$ and $\tilde{H}$ are only defined for $t \le \max(T_i) = t_{\max}$. If the largest observation is censored, $\hat{S}$ does not reach 0. Some extensions for $t > t_{\max}$:

– Efron (1967): $\hat{S}(t) = 0$;
– Gill (1980): $\hat{S}(t) = \hat{S}(t_{\max})$;
– Brown, Hollander & Kowar (1974): $\hat{S}(t) = \exp\{t \log[\hat{S}(t_{\max})]/t_{\max}\}$ (an exponential tail).

7 The 6-MP Leukaemia Trial

Data. 42 children with acute leukaemia in complete or partial remission were matched in pairs by remission status and randomized within the pair to 6-mercaptopurine (6-MP) or placebo maintenance therapy, then followed until relapse or end of study. Times to relapse (weeks) for the 21 **6-MP** patients ($^+$ = censored):

$$6,\ 6,\ 6,\ 6^+,\ 7,\ 9^+,\ 10,\ 10^+,\ 11^+,\ 13,\ 16,\ 17^+,\ 19^+,\ 20^+,\ 22,\ 23,\ 25^+,\ 32^+,\ 32^+,\ 34^+,\ 35^+.$$

Placebo times (all observed): $1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23$.

Example 5.8: Kaplan–Meier estimate for the 6-MP group

t_j	d_j	Y_j	$\hat{S}(t) = \prod(1 - d_j/Y_j)$	$\sum \frac{d_j}{Y_j(Y_j - d_j)}$	$\hat{S}(t)^2 \sum \cdots$
6	3	21	$[1 - \frac{3}{21}] = 0.857$	$\frac{3}{21\times 18} = 0.0079$	0.0058
7	1	17	$0.857(1 - \frac{1}{17}) = 0.807$	$0.0079 + \frac{1}{17\times 16} = 0.0116$	0.0076
10	1	15	$0.807(1 - \frac{1}{15}) = 0.753$	$0.0116 + \frac{1}{15\times 14} = 0.0164$	0.0093
13	1	12	$0.753(1 - \frac{1}{12}) = 0.690$	$0.0164 + \frac{1}{12\times 11} = 0.0240$	0.0114
16	1	11	$0.690(1 - \frac{1}{11}) = 0.628$	$0.0240 + \frac{1}{11\times 10} = 0.0330$	0.0130
22	1	7	$0.628(1 - \frac{1}{7}) = 0.538$	$0.0330 + \frac{1}{7\times 6} = 0.0569$	0.0164
23	1	6	$0.538(1 - \frac{1}{6}) = 0.448$	$0.0569 + \frac{1}{6\times 5} = 0.0902$	0.0181

Note how censorings reduce Y_j without producing a jump: e.g. between $t = 7$ ($Y = 17$) and $t = 10$ ($Y = 15$) the subject censored at 9^+ leaves the risk set.

Example 5.9: Nelson–Aalen estimate for the 6-MP group

t	$\tilde{H}(t) = \sum d_j/Y_j$	$\sigma_H^2 = \sum d_j/Y_j^2$	SE
$0 \le t < 6$	0	0	0
$6 \le t < 7$	$3/21 = 0.1428$	$3/21^2 = 0.0068$	0.0825
$7 \le t < 10$	$0.1428 + 1/17 = 0.2017$	$0.0068 + 1/17^2 = 0.0103$	0.1015
$10 \le t < 13$	$0.2017 + 1/15 = 0.2683$	$0.0103 + 1/15^2 = 0.0147$	0.1212
$13 \le t < 16$	$0.2683 + 1/12 = 0.3517$	$0.0147 + 1/12^2 = 0.0217$	0.1473
$16 \le t < 22$	$0.3517 + 1/11 = 0.4426$	$0.0217 + 1/11^2 = 0.0299$	0.1729
$22 \le t < 23$	$0.4426 + 1/7 = 0.5854$	$0.0299 + 1/7^2 = 0.0503$	0.2243
$23 \le t < 35$	$0.5854 + 1/6 = 0.7521$	$0.0503 + 1/6^2 = 0.0781$	0.2795

Comparison. Plotting $\tilde{S} = e^{-\tilde{H}}$ against $\hat{S}$ (K&M Fig. 4.1A), the Nelson–Aalen-based curve lies slightly *above* the KM curve at every step, and correspondingly $\tilde{H} = -\log \hat{S}$ lies slightly above $\tilde{H}$; the gap widens in the tail where Y_j is small (e.g. at $t = 23$: $\hat{S} = 0.448$ vs. $\tilde{S} = e^{-0.752} = 0.471$).

8 Selected exercises

Motivation. Three structural facts about the estimators of this lecture: without censoring they reduce to the classical empirical quantities, the Kaplan–Meier estimator redistributes each censored subject over its own future, and the Nelson–Aalen estimator matches the exact moments of the true cumulative hazard at the order statistics.

Exercise 5.1 (No censoring: KM and Greenwood reduce to binomial quantities)

Suppose no observation is censored ($\delta_i = 1$ for all i; ties allowed).
(a) Show that $\hat{S}(t) = 1 - \mathbb{F}_n(t)$ for all t.
(b) Show that Greenwood's formula reduces to $\hat{V}[\hat{S}(t)] = \hat{S}(t)\{1-\hat{S}(t)\}/n$ whenever $\hat{S}(t) > 0$.

Solution [Supplement].

(a)

Step 1. Put $Y_{D+1} = 0$. Without censoring, a subject with $T_i \geq t_j$ either dies at t_j or has $T_i \geq t_{j+1}$, so $Y_{j+1} = Y_j - d_j$ and $Y_1 = n$.

Step 2. Hence $1 - d_j/Y_j = Y_{j+1}/Y_j$, and for $t_k \leq t < t_{k+1}$ (with $t_{D+1} = \infty$) the product telescopes:

$$\hat{S}(t) = \prod_{j=1}^{k} \frac{Y_{j+1}}{Y_j} = \frac{Y_{k+1}}{n} = \frac{\#\{i : T_i > t\}}{n} = 1 - \mathbb{F}_n(t).$$

For $t < t_1$ both sides equal 1.

(b) If $\hat{S}(t) > 0$ then $Y_{k+1} > 0$, and each Greenwood term telescopes:

$$\frac{d_j}{Y_j(Y_j - d_j)} = \frac{Y_j - Y_{j+1}}{Y_j Y_{j+1}} = \frac{1}{Y_{j+1}} - \frac{1}{Y_j}, \qquad \sum_{j=1}^{k} \frac{d_j}{Y_j(Y_j - d_j)} = \frac{1}{Y_{k+1}} - \frac{1}{n}.$$

With $\hat{S} = Y_{k+1}/n$,

$$\hat{V}[\hat{S}(t)] = \frac{Y_{k+1}^2}{n^2}\Big(\frac{1}{Y_{k+1}} - \frac{1}{n}\Big) = \frac{1}{n} \cdot \frac{Y_{k+1}}{n}\Big(1 - \frac{Y_{k+1}}{n}\Big) = \frac{\hat{S}(t)\{1 - \hat{S}(t)\}}{n},$$

the usual binomial variance estimate for the proportion $\#\{T_i > t\}/n$. ■

Exercise 5.2 (Self-consistency of the Kaplan–Meier estimator)

Let t be such that $\hat{S}(t) > 0$. Show that

$$\hat{S}(t) = \frac{1}{n}\Big[\#\{i : T_i > t\} + \sum_{i:\, \delta_i = 0,\ T_i \leq t} \frac{\hat{S}(t)}{\hat{S}(T_i)}\Big].$$

That is, $\hat{S}(t)$ is the empirical survival fraction once every subject censored at $T_i \leq t$ is counted as alive at t with its own estimated probability $\hat{P}(X > t \mid X > T_i)$.

Solution [Supplement].

Step 1. For $s \leq t$ let $R(s) = \#\{i : T_i > s\}$ and $A(s) = \sum_{i:\delta_i=0, T_i \leq s} 1/\hat{S}(T_i)$; these are well defined because $\hat{S}$ is nonincreasing and $\hat{S}(s) \geq \hat{S}(t) > 0$. Put

$$\varphi(s) = \frac{R(s)}{\hat{S}(s)} + A(s).$$

The claim is $\varphi(t) = n$, multiplied through by $\hat{S}(t)/n$.

Step 2. R, $\hat{S}$ and A are right-continuous step functions that change only at observed times T_i. For $s < \min_i T_i$, $\varphi(s) = n/1 + 0 = n$.

Step 3. Let $s \le t$ be an observed time, $Y = \#\{i : T_i \ge s\} = R(s-)$, with d deaths and c censorings at s. Then $R(s) = Y - d - c$, $A(s) = A(s-) + c/\hat{S}(s)$ and $\hat{S}(s) = \hat{S}(s-)(1 - d/Y)$ (also when $d = 0$). Hence

$$\varphi(s) - \varphi(s-) = \frac{Y - d - c}{\hat{S}(s)} + \frac{c}{\hat{S}(s)} - \frac{Y}{\hat{S}(s-)} = \frac{(Y - d) - Y(1 - d/Y)}{\hat{S}(s)} = 0.$$

The tie convention matters here: the c subjects censored at s are in Y, which is what makes $\hat{S}(s) = \hat{S}(s-)(1 - d/Y)$.

Step 4. By Steps 2–3, φ is constant, equal to n, on $(-\infty, t]$, which proves the identity.

Efron (1967) showed that this "redistribute-to-the-right" property characterizes $\hat{S}$; it is the template for Turnbull's algorithms in Lecture 8. ■

Exercise 5.3 (Nelson–Aalen at the order statistics)

Let $X_1, \dots, X_n$ be i.i.d. with continuous F, no censoring, and order statistics $X_{(1)} < \cdots < X_{(n)}$.
(a) Show that $\tilde{H}(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{n-i+1}$ and $\sigma_H^2(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{(n-i+1)^2}$.
(b) Show that $\mathrm{E}\{H(X_{(k)})\} = \tilde{H}(X_{(k)})$ and $\mathrm{Var}\{H(X_{(k)})\} = \sigma_H^2(X_{(k)})$.

Solution [Supplement].

(a) Continuity of F excludes ties a.s., so at $t_i = X_{(i)}$ we have $d_i = 1$ and $Y_i = n - i + 1$. Substitute into Theorem 5.5.

(b)

Step 1. Since F is continuous, $S(X) = 1 - F(X) \sim U(0,1)$, so for $u \ge 0$, $P\{H(X) > u\} = P\{S(X) < e^{-u}\} = e^{-u}$: the $E_i = H(X_i)$ are i.i.d. Exp(1).

Step 2. H is nondecreasing, so it preserves order: $H(X_{(k)}) = E_{(k)}$, the k-th order statistic of $E_1, \dots, E_n$.

Step 3. The joint density of $(E_{(1)}, \dots, E_{(n)})$ is $n! \exp(-\sum_i e_i)$ on $0 < e_1 < \cdots < e_n$. Let $D_i = E_{(i)} - E_{(i-1)}$, $E_{(0)} = 0$; this linear map is triangular with unit diagonal, so its Jacobian is 1, and $\sum_i e_i = \sum_i (n - i + 1) d_i$. Since $\prod_{i=1}^{n} (n - i + 1) = n!$, the density of $(D_1, \dots, D_n)$ is

$$\prod_{i=1}^{n} (n - i + 1)\, e^{-(n-i+1)d_i}, \qquad d_i > 0,$$

so the spacings are independent with $D_i \sim \mathrm{Exp}(n - i + 1)$.

Step 4. $E_{(k)} = \sum_{i=1}^{k} D_i$, hence $\mathrm{E}E_{(k)} = \sum_{i \le k} (n - i + 1)^{-1}$ and, by independence, $\mathrm{Var}E_{(k)} = \sum_{i \le k} (n - i + 1)^{-2}$. Compare with (a).

So the Nelson–Aalen estimate at $X_{(k)}$ is exactly the mean of the true cumulative hazard there, and Aalen's variance is exactly its variance. ■

Lecture 6. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 2)

1 Recap

$$\hat{S}(t) = \prod_{t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}; \qquad \tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j}, \quad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Throughout, define

$$\sigma_S^2(t) = \frac{\hat{V}[\hat{S}(t)]}{\hat{S}^2(t)} = \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

so that $\hat{S}(t)\sigma_S(t)$ is the Greenwood standard error.

2 Pointwise Confidence Intervals for $S(t)$

Motivation. An estimate of $S(t)$ needs an interval. Greenwood's variance gives one directly, but the interval may leave $[0, 1]$; transformations such as log and log-log fix this.

2.1 Linear interval

Definition 6.1: Linear CI

$$\Big[\hat{S}(t) - Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t), \quad \hat{S}(t) + Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t)\Big].$$

It is symmetric about $\hat{S}(t)$ and can fall outside $[0, 1]$.

2.2 Transformed intervals: the general recipe

Derivation: delta method for a transformation g

Let g be known, monotone and differentiable. By the delta method, $\mathrm{SE}\{g(\hat{S})\} \approx |g'(\hat{S})| \cdot \mathrm{SE}(\hat{S}) = |g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)$. A $100(1-\alpha)\%$ CI for $g(S(t))$ is

$$g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t),$$

and re-transforming gives a CI for $S(t)$:

$$g^{-1}\Big\{g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)\Big\}.$$

Why transform? Convergence to normality may be faster on the transformed scale, and the re-transformed CI respects $0 \le S(t) \le 1$.

2.3 Log-log transformation

Theorem 6.2: Log-log CI

$$\Big(\hat{S}(t)^{1/\theta},\ \hat{S}(t)^{\theta}\Big), \qquad \theta = \exp\Big\{\frac{Z_{1-\alpha/2}\,\sigma_S(t)}{\log \hat{S}(t)}\Big\}.$$

Derivation [Supplement]. K&M use $\log H(t) = \log\{-\log S(t)\}$, i.e. $g(u) = \log\{-\log u\}$ with $g'(u) = \dfrac{1}{u \log u}$ and $g^{-1}(v) = \exp\{-\exp(v)\}$. Then

$$|g'(\hat{S})|\hat{S}\sigma_S = \frac{\sigma_S}{|\log \hat{S}|},$$

so the CI for $g(S)$ is $\log\{-\log \hat{S}\} \pm a$ with $a = Z_{1-\alpha/2}\sigma_S/|\log \hat{S}|$. Re-transforming,

$$g^{-1}\big(\log\{-\log \hat{S}\} \pm a\big) = \exp\{\log \hat{S} \cdot e^{\pm a}\} = \hat{S}^{e^{\pm a}}.$$

Since $\log \hat{S} < 0$, $\theta = e^{-a} < 1$; the endpoints are $\hat{S}^{1/\theta}$ (lower, as $1/\theta > 1$) and $\hat{S}^{\theta}$ (upper). ■

2.4 Arcsine-square-root transformation

Theorem 6.3: Arcsine-square-root CI

$$\text{lower} = \sin^2\Big\{\max\Big[0,\ \arcsin\big(\hat{S}(t)^{1/2}\big) - 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\},$$

$$\text{upper} = \sin^2\Big\{\min\Big[\frac{\pi}{2},\ \arcsin\big(\hat{S}(t)^{1/2}\big) + 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\}.$$

Derivation [Supplement]. $g(u) = \arcsin(\sqrt{u})$ has $g'(u) = \dfrac{1}{\sqrt{1-u}} \cdot \dfrac{1}{2\sqrt{u}}$, so $|g'(\hat{S})|\hat{S}\sigma_S = \dfrac{\hat{S}\sigma_S}{2\sqrt{\hat{S}(1-\hat{S})}} = 0.5\,\sigma_S\Big(\dfrac{\hat{S}}{1-\hat{S}}\Big)^{1/2}$. The max / min keep the angle in $[0, \pi/2]$, and $g^{-1}(v) = \sin^2 v$. ■

2.5 Log transformation

Definition 6.4: Log CI (default of R `survfit`)

$g(S) = \log S$, $g'(u) = 1/u$, so $|g'(\hat{S})|\hat{S}\sigma_S = \sigma_S$:

$$\min\Big(\exp\big\{\log \hat{S}(t) \pm Z_{1-\alpha/2}\sigma_S(t)\big\},\ 1\Big).$$

Less recommended than log-log and arcsine-square-root.

Remark 6.5: Comments

- Log-log and arcsine-square-root CIs have better coverage than the linear CI.
- For very large samples the three methods are equivalent.

- The transformed CIs are *not* symmetric about the point estimate; they are appropriate in small samples where the point estimator is biased and its distribution skewed.

Example 6.6: Toy example, $n = 7$ [Supplement]

T_i	1.5	2.5	1.4	6.2	2.8	5.3	4.5
δ_i	1	0	0	1	1	1	0

Ordered: 1.4^+, 1.5, 2.5^+, 2.8, 4.5^+, 5.3, 6.2.

t_j	d_j	Y_j	$\hat{S}$	σ_S	Linear	Log-log	Arcsine
1.5	1	6	0.833	0.183	(0.535, 1.000*)	(0.273, 0.975)	(0.465, 1.000)
2.8	1	4	0.625	0.342	(0.207, 1.000*)	(0.142, 0.893)	(0.213, 0.949)
5.3	1	2	0.313	0.785	(0.000*, 0.794)	(0.013, 0.734)	(0.006, 0.804)
6.2	1	1	0	—	—	—	—

*The raw linear limits (1.131, 1.043, −0.169) fall outside $[0, 1]$; transformed intervals stay inside automatically. Computation at $t = 2.8$: $\sigma_S^2 = \frac{1}{6\cdot5} + \frac{1}{4\cdot3} = 0.1167$, $\theta = \exp\{1.96 \times 0.3416/\ln 0.625\} = 0.241$, so the log-log CI is $(0.625^{1/0.241}, 0.625^{0.241}) = (0.142, 0.893)$.

3 Pointwise Confidence Intervals for $H(t)$

Based on the Nelson–Aalen $\tilde{H}(t)$ and $\sigma_H(t)$:

- **Linear**: $\tilde{H}(t) \pm Z_{1-\alpha/2}\sigma_H(t)$.
- **Log-transformed** (on H): $[\tilde{H}(t)/\phi,\ \phi\tilde{H}(t)]$, $\phi = \exp\left[\dfrac{Z_{1-\alpha/2}\sigma_H(t)}{\tilde{H}(t)}\right]$ (since $\mathrm{SE}(\log\tilde{H}) \approx \sigma_H/\tilde{H}$).
- **Arcsine-square-root** (on $S = e^{-H}$):

$$\Big[-2\ln\Big\{\sin\Big(\min\Big[\frac{\pi}{2},\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) + 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\},$$
$$-2\ln\Big\{\sin\Big(\max\Big[0,\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) - 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\}\Big].$$

4 Confidence Bands

4.1 Pointwise intervals vs. bands

For a fixed t_0, a $100(1-\alpha)\%$ CI $[L_{t_0}, U_{t_0}]$ satisfies $P\{L_{t_0} \le S(t_0) \le U_{t_0}\} = 1-\alpha$; once t_0 is fixed, $S(t_0)$ is just a scalar. Hence:

- a plot of pointwise intervals over all t should *not* be read as an interval for the entire survival curve;
- such a plot is always "too narrow" compared with a band for the whole $S(t)$.

Definition 6.7: Confidence band

$[L(t), U(t)]$ is a $100(1-\alpha)\%$ confidence band (CB) for $S(t)$ on $t_L \le t \le t_U$ if

$$P\{L(t) \le S(t) \le U(t) \text{ for all } t_L \le t \le t_U\} = 1-\alpha,$$

where $t_L \ge$ the smallest and $t_U \le$ the largest observed event time.

4.2 Equal probability (EP) bands

- Define
$$a_L = \frac{n\sigma_S^2(t_L)}{1+n\sigma_S^2(t_L)}, \qquad a_U = \frac{n\sigma_S^2(t_U)}{1+n\sigma_S^2(t_U)}, \qquad 0 < a_L \le a_U < 1.$$
- EP bands are **proportional to the pointwise confidence bounds**.
- From Table C.3 (K&M Appendix C) find the coefficient $c_\alpha(a_L, a_U)$ and build the band over $[t_L, t_U]$ by replacing $Z_{1-\alpha/2}$ in Definition 6.1 and Theorems 6.2–6.3 with $c_\alpha(a_L, a_U)$. (E.g. for a 90% band, $c_{0.10}(0.10, 0.40) = 2.4721$, larger than $Z_{0.95} = 1.645$.)

4.3 Hall–Wellner (HW) bands

- **Not** proportional to the pointwise bounds: wider for small t and narrower for large t.
- Find $k_\alpha(a_L, a_U)$ in Table C.4 and replace $Z_{1-\alpha/2}\sigma_S(t)$ in Definition 6.1 and Theorems 6.2–6.3 with
$$k_\alpha(a_L, a_U)\left[1+n\sigma_S^2(t)\right] n^{-1/2}.$$

4.4 Performance and sample-size advice

- The EP *linear* band performs poorly when $n < 200$: coverage is considerably below nominal.
- Log-log and arcsine-square-root EP bands have approximately correct coverage for $n \ge 20$; the arcsine EP band seems slightly better than the log-log EP band.
- For HW bands, all three versions perform reasonably well with as few as 20 observed events.

EP	Sample size	HW	Sample size
Linear	≥ 200	Linear	≥ 20
Log-log	≥ 20	Log-log	≥ 20
Arcsine	≥ 20	Arcsine	≥ 20

Example 6.8: Bone marrow transplant, ALL patients (K&M Table 4.3)

For 38 ALL patients, the KM estimate drops from $\hat{S}(1) = 0.9737$ ($Y = 38$) to $\hat{S}(662) = 0.3531$ ($Y = 13$), staying flat thereafter to the last (censored) time 2081; the Nelson–Aalen estimate reaches $\tilde{H}(662) = 1.0152$ with $\sigma_H = 0.2185$. Comparing 95% pointwise CIs, the EP band and the HW band for disease-free survival on roughly 100–600 days (K&M Fig. 4.5): both bands are wider than the pointwise intervals, and the HW band is wider than the EP band at early times but narrower later, as described above.

5 Selected exercises

Motivation. The interval formulas of this lecture are linked: the log interval for H is the log-log interval for S in disguise, the arcsine interval for H is the arcsine interval for S read through $H = -\log S$, and without censoring the band constants reduce to familiar empirical-process objects.

Exercise 6.1 (Log interval for H versus log-log interval for S)

Let $\tilde{H}(t) > 0$ and $\tilde{S}(t) = e^{-\tilde{H}(t)}$.
(a) Show that mapping the log-transformed CI for $H(t)$ through $S = e^{-H}$ gives $(\tilde{S}(t)^{\phi}, \tilde{S}(t)^{1/\phi})$, which is the log-log CI of Theorem 6.2 with $\hat{S}$ replaced by $\tilde{S}$ and σ_S by σ_H.
(b) Show that $\sigma_H^2(t) \le \sigma_S^2(t)$ when all $Y_j > d_j$ ($t_j \le t$), with

$$\sigma_S^2(t) - \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j^2}{Y_j^2(Y_j - d_j)}.$$

Solution [Supplement].

(a)

Step 1. $H \mapsto e^{-H}$ is strictly decreasing, so the image of $[\tilde{H}/\phi, \phi\tilde{H}]$ is $[e^{-\phi\tilde{H}}, e^{-\tilde{H}/\phi}] = [\tilde{S}^{\phi}, \tilde{S}^{1/\phi}]$.

Step 2. With $\hat{S} \to \tilde{S}$ and $\sigma_S \to \sigma_H$, the constant of Theorem 6.2 is

$$\theta = \exp\Big\{\frac{Z_{1-\alpha/2}\sigma_H}{\log \tilde{S}}\Big\} = \exp\Big\{-\frac{Z_{1-\alpha/2}\sigma_H}{\tilde{H}}\Big\} = \frac{1}{\phi}.$$

Hence $(\tilde{S}^{1/\theta}, \tilde{S}^{\theta}) = (\tilde{S}^{\phi}, \tilde{S}^{1/\phi})$, the interval of Step 1. Both are Wald intervals on the scale $\log H = \log(-\log S)$.

(b) Termwise,

$$\frac{d_j}{Y_j(Y_j - d_j)} - \frac{d_j}{Y_j^2} = \frac{d_j\{Y_j - (Y_j - d_j)\}}{Y_j^2(Y_j - d_j)} = \frac{d_j^2}{Y_j^2(Y_j - d_j)} \ge 0;$$

sum over $t_j \le t$. The relative difference of the j-th terms is $d_j/(Y_j - d_j)$, negligible unless the risk set is small, so the two intervals in (a) differ mainly in the tail. ■

Exercise 6.2 (Derivation of the arcsine-square-root interval for H)

Derive the arcsine-square-root CI for $H(t)$ stated in this lecture from Theorem 6.3.

Solution [Supplement].

Step 1 (standard error on the S scale). By the delta method for $\tilde{S} = e^{-\tilde{H}}$, $\mathrm{SE}(\tilde{S}) \approx \tilde{S}\,\sigma_H$. So σ_H plays the role of $\sigma_S = \mathrm{SE}(\hat{S})/\hat{S}$, and Theorem 6.3 applies with $\hat{S} \to \tilde{S}$, $\sigma_S \to \sigma_H$.

Step 2 (rewrite in $\tilde{H}$). $\arcsin(\tilde{S}^{1/2}) = \arcsin(e^{-\tilde{H}/2})$ and

$$\Big(\frac{\tilde{S}}{1-\tilde{S}}\Big)^{1/2} = \Big(\frac{e^{-\tilde{H}}}{1-e^{-\tilde{H}}}\Big)^{1/2} = (e^{\tilde{H}} - 1)^{-1/2}.$$

So the angle interval is $[v_-, v_+]$ with

$$v_{\pm} = \mathrm{clip}_{[0,\pi/2]}\Big\{\arcsin\left(e^{-\tilde{H}/2}\right) \pm 0.5 Z_{1-\alpha/2}\sigma_H (e^{\tilde{H}} - 1)^{-1/2}\Big\},$$

and the CI for S is $[\sin^2 v_-, \sin^2 v_+]$.

Step 3 (back to H). $H = -\log S = -2\log\sin v$ is strictly decreasing in $v \in (0, \pi/2]$, so the endpoints swap: the CI for H is $[-2\log\sin v_+,\ -2\log\sin v_-]$, which is the stated formula (the $\min[\pi/2, \cdot]$ goes with the lower limit and the $\max[0, \cdot]$ with the upper). If $v_- = 0$ the upper limit is $+\infty$. ■

Exercise 6.3 (Band constants without censoring)

Let $a(t) = n\sigma_S^2(t)/\{1 + n\sigma_S^2(t)\}$, so that $a_L = a(t_L)$ and $a_U = a(t_U)$.
(a) Show that a is nondecreasing in t with values in $[0, 1)$.
(b) Without censoring, show that $a(t) = 1 - \hat{S}(t) = \mathbb{F}_n(t)$ whenever $\hat{S}(t) > 0$.
(c) Without censoring, show that the linear HW band is $\hat{S}(t) \pm k_\alpha(a_L, a_U)\, n^{-1/2}$, a band of constant width.

Solution [Supplement].

(a) $\sigma_S^2(t)$ is a sum of nonnegative terms over $t_j \le t$, hence nondecreasing in t, and $x \mapsto x/(1+x)$ is increasing from $[0, \infty)$ onto $[0, 1)$.

(b) By Exercise 5.1(b), $\sigma_S^2(t) = \{1 - \hat{S}(t)\}/\{n\hat{S}(t)\}$, so $1 + n\sigma_S^2 = 1/\hat{S}$ and

$$a(t) = \frac{(1 - \hat{S})/\hat{S}}{1/\hat{S}} = 1 - \hat{S}(t) = \mathbb{F}_n(t)$$

by Exercise 5.1(a). So $[a_L, a_U]$ is the range of the empirical cdf over $[t_L, t_U]$, the natural time scale for the Brownian-bridge limit of $\sqrt{n}(\mathbb{F}_n - F)$.

(c) The HW band replaces $Z_{1-\alpha/2}\sigma_S(t)$ in Definition 6.1 by $k_\alpha[1 + n\sigma_S^2(t)]n^{-1/2} = k_\alpha n^{-1/2}/\hat{S}(t)$ (by (b)). The half-width $\hat{S}(t) \cdot k_\alpha n^{-1/2}/\hat{S}(t) = k_\alpha n^{-1/2}$ does not depend on t: a Kolmogorov–Smirnov-type band. ■

Lecture 7. Estimation of Basic Quantities for Other Sampling Schemes

1 Estimating the Mean Survival Time

1.1 Mean as the area under the curve

Since $\mu = \mathrm{E}(X) = \int_0^\infty S(t)\,\mathrm{d}t$, the natural estimator is $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. Problem: if the last observation is censored, $\hat{S}(t)$ stays flat (never reaches 0) and the integral diverges. The right tail can be completed, but the estimate of the mean then depends on which completion is used.

1.2 First solution: Efron's completion

Setting $\hat{S}(t) = 0$ for $t > t_{\max}$ (Efron) gives $\hat{\mu} = \int_0^{t_{\max}} \hat{S}(t)\,\mathrm{d}t$. Downsides: it assumes all survivors at $t_{\max}$ die immediately afterwards (biased downward), and $t_{\max}$ varies from sample to sample.

1.3 Second solution: restricted mean

Theorem 7.1: Restricted mean

For a pre-specified τ, estimate $\mu_\tau = \mathrm{E}[\min(X,\tau)] = \int_0^\tau S(t)\,\mathrm{d}t$ by

$$\hat{\mu}_\tau = \int_0^\tau \hat{S}(t)\,\mathrm{d}t, \qquad \hat{V}(\hat{\mu}_\tau) = \sum_{j=1}^{D} \Big[\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t\Big]^2 \frac{d_j}{Y_j(Y_j - d_j)},$$

with $100(1-\alpha)\%$ CI $\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$. (The sum runs over death times $t_j \le \tau$.)

Remark 7.2: Why the weights $\int_{t_j}^\tau \hat{S}$

A perturbation of the hazard estimate at t_j rescales $\hat{S}(t)$ for every $t \ge t_j$, hence changes the area to the right of t_j only. By the delta method, the contribution of $\hat{h}_j$ to $\mathrm{Var}(\hat{\mu}_\tau)$ is (area right of t_j)$^2\times$ Greenwood's term $d_j/[Y_j(Y_j - d_j)]$.

Since $\hat{S}$ is a step function, $\hat{\mu}_\tau$ is a sum of rectangles: $\hat{\mu}_\tau = \sum_k \hat{S}(s_k)(s_{k+1} - s_k)$ over consecutive jump points s_k in $[0,\tau]$.

Example 7.3: 6-MP group (data of Lecture 5) [Supplement]

KM: $\hat{S} = 0.857, 0.807, 0.753, 0.690, 0.627, 0.538, 0.448$ at $t_j = 6, 7, 10, 13, 16, 22, 23$ ($Y_j = 21, 17, 15, 12, 11, 7, 6$). With $\tau = 35$ (the largest, censored, time):

$$\begin{aligned}\hat{\mu}_{35} &= 6(1) + 1(0.857) + 3(0.807) + 3(0.753) + 3(0.690)\\ &\quad + 6(0.627) + 1(0.538) + 12(0.448) \approx 23.29 \text{ weeks},\\ \hat{V}(\hat{\mu}_{35}) &\approx 7.99, \quad \mathrm{SE} \approx 2.83, \quad 95\%\ \mathrm{CI} \approx (17.75,\ 28.83).\end{aligned}$$

2 Quantiles and Their Confidence Intervals

Recall $x_p = \inf\{t : S(t) \le 1-p\}$ ($p = 0.5$: median). The estimator is $\hat{x}_p = \inf\{t : \hat{S}(t) \le 1-p\}$; its variance depends on the density $f(x_p)$, which is hard to estimate. Instead, CIs are obtained by **inverting** the pointwise CIs for $S(t)$.

Theorem 7.4: CIs for x_p

A $100(1-\alpha)\%$ CI for x_p is the set of all t satisfying

$$\text{linear:} \quad -Z_{1-\alpha/2} \le \frac{\hat{S}(t)-(1-p)}{\hat{V}^{1/2}[\hat{S}(t)]} \le Z_{1-\alpha/2},$$

$$\text{log-log:} \quad \left|\frac{[\log\{-\log\hat{S}(t)\}-\log\{-\log(1-p)\}]\hat{S}(t)\log\hat{S}(t)}{\hat{V}^{1/2}[\hat{S}(t)]}\right| \le Z_{1-\alpha/2},$$

$$\text{arcsine:} \quad \left|\frac{2\{\arcsin\sqrt{\hat{S}(t)}-\arcsin\sqrt{1-p}\}\{\hat{S}(t)(1-\hat{S}(t))\}^{1/2}}{\hat{V}^{1/2}[\hat{S}(t)]}\right| \le Z_{1-\alpha/2}.$$

Derivation: where these come from

Each is a Wald test of $H_0 : S(t) = 1-p$ on the transformed scale: $|g(\hat{S}) - g(1-p)|/(|g'(\hat{S})|\,\mathrm{SE}(\hat{S})) \le Z$. For log-log, $1/|g'(\hat{S})| = |\hat{S}\log\hat{S}|$; for arcsine-square-root, $1/|g'(\hat{S})| = 2\sqrt{\hat{S}(1-\hat{S})}$. The CI collects all t at which H_0 is not rejected.

Example 7.5: Median for the 6-MP group

$\hat{S}(22) = 0.538 > 0.5 \ge \hat{S}(23) = 0.448$, so $\hat{x}_{0.5} = 23$. Checking the criteria at each death time, all three methods are satisfied from $t = 13$ onwards ($\hat{S}(13) = 0.690$, SE 0.107) but not at $t \le 10$. Since $\hat{S}$ never drops far enough below 0.5 to violate the criterion again, the 95% CI for the median is $[13, \infty)$: the upper limit cannot be determined.

3 Right-Censored and Left-Truncated Data

Definition 7.6: Data and modified risk set

Data $\{(L_i, T_i, \delta_i)\}$: L_i = age at entering the study, T_i = death or censoring time. At death times $t_1 < \cdots < t_D$ let d_j = number of deaths and

$$Y_j = \sum_{i=1}^{n} I(L_i \le t_j \le T_i),$$

the number who have *entered* and are still under observation at t_j.

With this modified Y_j, all estimators of K&M Sections 4.2–4.4 (KM, Nelson–Aalen, Greenwood, CIs, bands) apply unchanged.

Remark 7.7: What is being estimated

Subjects are only seen after their entry time, so the estimators target *conditional* quantities:

$$S_L(t) = P(X > t \mid X > L), \qquad H_L(t) = -\ln P(X > t) + \ln P(X > L), \qquad H_L'(t) = h(t).$$

The hazard itself is unaffected by conditioning, which is why d_j/Y_j still estimates it.

Example 7.8: Channing House

Residents of the Channing House retirement center (California): 97 males and 365 females; truncation times are the ages (months) at entry. The number at risk is small at young ages, rises to a peak around 950–1000 months, then falls.

- Males: the first subject entered at 751 and died at 777; the second entered at 759 and died at 781; the third appeared only at 782.
- Plain KM conditional on the youngest entry: at 777, $Y = 2$, $d = 1 \Rightarrow$ factor $1/2$; at 781, $Y = 1$, $d = 1 \Rightarrow$ factor 0. So $\hat{S}_L(t) = 0$ for all $t \geq 781$ — the estimate is destroyed by the tiny early risk set.
- Remedy: estimate $S_a(t) = P(X > t \mid X \geq a)$ for a chosen age a where the risk set is reasonably large:

$$\hat{S}_a(t) = \prod_{a \leq t_j \leq t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}_a(t)] = \hat{S}_a(t)^2 \sum_{a \leq t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)},$$

$$\hat{H}_a(t) = \sum_{a \leq t_j \leq t} \frac{d_j}{Y_j}, \qquad t \geq a.$$

4 Left-Censored Data: Reversing the Time Scale

X = failure time, C = left censoring time; observe $T = \max(X, C)$, $\delta = I(T = X) = I(X \geq C)$. How to estimate $S(u) = P(X > u)$?

Theorem 7.9: Reversed time-scale trick

Pick $\tau > \max_i T_i$. Then $\tau - X_i$ is *right* censored by $\tau - C_i$:

$$\tau - T_i = \min(\tau - X_i,\ \tau - C_i), \qquad \delta_i = I(\tau - X_i \leq \tau - C_i).$$

Let $G(u) = P(\tau - X > u)$. Apply the usual KM to $\{(\tau - T_i, \delta_i)\}$ to get $\hat{G}$, and set

$$\hat{S}(u) = 1 - \hat{G}(\tau - u), \qquad \mathrm{Var}[\hat{S}(u)] = \mathrm{Var}[\hat{G}(\tau - u)].$$

CIs and bands are built the same way.

Why [Supplement]. $\max(X, C) = \tau - \min(\tau - X, \tau - C)$, and $X \geq C \iff \tau - X \leq \tau - C$. Also $G(\tau - u) = P(\tau - X > \tau - u) = P(X < u)$, so $S(u) = 1 - G(\tau - u)$ (for continuous X). ■

5 Right-Truncated Data

We observe $\{(R_i, X_i)\}$ sampled conditionally on $X_i < R_i$. Reverse the time scale with $\tau > \max_i R_i$: then $\tau - X_i$ is *left truncated* by $\tau - R_i$ (since $X_i < R_i \iff \tau - X_i > \tau - R_i$). Use the standard left-truncation method to estimate the conditional distribution of $\tau - X_i$ given $\tau - X_i > a$.

Definition 7.10: Choice $a = \tau - \tau^*$, $\tau^* = \max_i R_i$

$$G^*(t) = P(\tau - X_i \geq t \mid \tau - X_i \geq \tau - \tau^*) = P(\tau - X_i \geq t \mid X_i \leq \tau^*),$$
$$S^*(t) = P(X_i > t \mid X_i \leq \tau^*) = 1 - G^*(\tau - t).$$

Theorem 7.11: KM on the reversed scale

With $\tau - t_j$ the D distinct failure times on the reversed scale,

$$\hat{G}^*(t) = \prod_{\tau - \tau^* \leq \tau - t_j \leq t} \left(1 - \frac{d_j}{Y_j}\right),$$

$$d_j = \#\{i : \tau - X_i = \tau - t_j\}, \quad Y_j = \#\{i : \tau - X_i \geq \tau - t_j \geq \tau - R_i\},$$

and $\hat{S}^*(t) = 1 - \hat{G}^*(\tau - t)$. Inference for S^* uses the properties of $\hat{G}^*$ for LT data.

Only the conditional survival given $X \leq \tau^*$ is estimable: nothing in the data speaks about $X > \tau^*$.

Remark 7.12: Quick review

KM and NA were developed for a single, possibly right-censored failure time with independent censoring. They extend directly to LT + RC data (modified risk set), and to left-censored or right-truncated data by reversing the time scale.

6 Other Sampling Schemes

Interval censoring, double censoring, and their combinations with left truncation have no simple closed-form estimator; Turnbull's iterative methods for these cases are covered in Lecture 8.

7 Selected exercises

Motivation. The restricted mean and the left-truncation risk set were used above without full justification; these exercises supply the identity behind μ_τ, the delta-method derivation of its variance, and the reason why d_j/Y_j with the modified risk set still estimates the hazard.

Exercise 7.1 (The restricted mean)

Let $X \geq 0$ have survival function S and let $\tau > 0$.
(a) Show that $\mathrm{E}[\min(X, \tau)] = \int_0^\tau S(t)\,\mathrm{d}t$.
(b) Without censoring and with $\tau \geq X_{(n)}$, show that $\hat{\mu}_\tau = \bar{X}$.

Solution [Supplement].

(a) Write $\min(x,\tau) = \int_0^\tau I(t<x)\,\mathrm{d}t$ for $x \geq 0$. By Tonelli (the integrand is nonnegative),

$$\mathrm{E}[\min(X,\tau)] = \int_0^\tau P(X>t)\,\mathrm{d}t = \int_0^\tau S(t)\,\mathrm{d}t.$$

(b) By Exercise 5.1(a), $\hat{S}(t) = n^{-1}\sum_i I(X_i > t)$, which vanishes for $t \geq X_{(n)}$. Hence

$$\hat{\mu}_\tau = \frac{1}{n}\sum_{i=1}^n \int_0^\tau I(t<X_i)\,\mathrm{d}t = \frac{1}{n}\sum_{i=1}^n \min(X_i,\tau) = \bar{X}.$$

So the restricted mean is the natural censored-data version of the sample mean. ■

Exercise 7.2 (Variance of the restricted mean)

Derive the variance estimator $\hat{V}(\hat{\mu}_\tau)$ of this lecture by the delta method, under the same approximations as Greenwood's formula (Lecture 5): the $\hat{h}_j = d_j/Y_j$ are approximately uncorrelated with $\mathrm{Var}(\hat{h}_j) \approx \hat{h}_j(1-\hat{h}_j)/Y_j$; assume $Y_j > d_j$ for $t_j \leq \tau$.

Solution [Supplement].

Step 1 (dependence on $\hat{h}_j$). $\hat{S}(t) = \prod_{t_j \leq t}(1-\hat{h}_j)$, so for $t_j \leq \tau$,

$$\frac{\partial \hat{S}(t)}{\partial \hat{h}_j} = -\frac{\hat{S}(t)}{1-\hat{h}_j}\,I(t \geq t_j).$$

Step 2 (gradient of $\hat{\mu}_\tau$). Differentiating under the (finite-sum) integral,

$$\frac{\partial \hat{\mu}_\tau}{\partial \hat{h}_j} = -\frac{1}{1-\hat{h}_j}\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t =: -\frac{A_j}{1-\hat{h}_j}.$$

Step 3 (delta method). With uncorrelated $\hat{h}_j$,

$$\mathrm{Var}(\hat{\mu}_\tau) \approx \sum_{t_j \leq \tau} \frac{A_j^2}{(1-\hat{h}_j)^2}\cdot\frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j\leq\tau} A_j^2 \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j\leq\tau} A_j^2 \frac{d_j}{Y_j(Y_j-d_j)},$$

which is $\hat{V}(\hat{\mu}_\tau)$. The same computation with the functional $\hat{S}(t)$ in place of $\hat{\mu}_\tau$ (gradient $-\hat{S}(t)/(1-\hat{h}_j)$) gives Greenwood's formula; for the area, $\hat{S}(t)$ is replaced by the area A_j to the right of t_j, as described in the Remark above. ■

Exercise 7.3 (Why the modified risk set works under left truncation)

Let X be discrete with support $\{x_j\}$ and hazard $h(x_j) = P(X=x_j)/P(X \geq x_j)$. Let the entry time L and censoring time C be such that X, L, C are mutually independent; a subject is sampled only if $L \leq X$ and then yields (L, T, δ) with $T = \min(X,C)$, $\delta = I(X \leq C)$. Show that, for every x_j with $P(L \leq x_j \leq T) > 0$,

$$P(T = x_j,\ \delta = 1 \mid L \leq x_j \leq T,\ \text{sampled}) = h(x_j).$$

Solution [Supplement].

Step 1 (sampling drops out). $\{L \leq x_j \leq T\}$ implies $L \leq x_j \leq X$, so it is contained in the sampling event $\{L \leq X\}$. Hence conditioning on "sampled" in addition changes nothing:

$$P(\cdot \mid L \leq x_j \leq T,\ \text{sampled}) = P(\cdot \mid L \leq x_j \leq T).$$

Step 2 (independence).

$$\frac{P(X = x_j,\ C \geq x_j,\ L \leq x_j)}{P(X \geq x_j,\ C \geq x_j,\ L \leq x_j)} = \frac{P(X = x_j)P(C \geq x_j)P(L \leq x_j)}{P(X \geq x_j)P(C \geq x_j)P(L \leq x_j)} = h(x_j).$$

The numerator is $P(T = x_j, \delta = 1,\ L \leq x_j \leq T)$, and the denominator is $P(L \leq x_j \leq T)$.

So among subjects with $L_i \leq t_j \leq T_i$ (exactly the modified risk set Y_j) the chance of an observed death at t_j is $h(t_j)$, and d_j/Y_j estimates it. The product of $1-h$ over $x_j \geq a$ gives $P(X > t \mid X \geq a)$, which is why only conditional survival is estimable. ■

Lecture 8. Estimation of Basic Quantities for Other Sampling Schemes (Part 2)

1 Other Sampling Schemes

- Interval censoring: $X_i \in (L_i, R_i]$, never observed exactly.
- Double censoring: each observation is LC, RC, or exact.
- LT + interval censoring, or LT + double censoring.

Estimation here is hard: there is no simple closed-form estimator, and inference is not easy.

2 Turnbull's Iterative Method (1974) for Double Censoring

Pure left censoring is rare; samples with both left and right censoring are more common. Turnbull (1974) proposed a modified product-limit estimator with no closed form, computed iteratively.

- **Idea.** Start with an estimate of S that ignores the left-censored data; use it to compute the *expected* number of deaths at each observed time contributed by the left-censored observations; update S; repeat until it stabilizes (a self-consistency / EM-type algorithm).
- **Notation.** $0 = t_0 < t_1 < \cdots < t_m$ the distinct observed times; d_j = deaths at t_j (possibly 0); c_j = left-censored observations at t_j; r_j = right-censored observations at t_j.

Definition 8.1: Turnbull (1974) algorithm

Step 0. Initial estimate $S_0(t_j)$, e.g. the KM ignoring the left-censored observations.

Step $(K{+}1).1$**.** For each left-censoring time t_i and $j \le i$, estimate $p_{ij} = P(t_{j-1} < X \le t_j \mid X \le t_i)$ by

$$p_{ij} = \frac{S_K(t_{j-1}) - S_K(t_j)}{1 - S_K(t_i)}.$$

Step $(K{+}1).2$**.** Estimated events at t_j: $\hat{d}_j = d_j + \sum_{i=j}^{m} c_i\, p_{ij}$ (each left-censored observation is distributed over the times before it).

Step $(K{+}1).3$**.** Compute the usual KM from the pseudo right-censored data with $\hat{d}_j$ events and r_j right-censored observations at t_j, ignoring the left-censored data: $Y_j = \sum_{l=j}^{m} (\hat{d}_l + r_l)$.

Repeat until $S_{K+1}(t_j) \approx S_K(t_j)$ for all t_j.

Remark 8.2: Why p_{ij} has this form

A boy left censored at t_i is known to have $X \le t_i$. Given that, the chance his event fell in $(t_{j-1}, t_j]$ is $P(t_{j-1} < X \le t_j)/P(X \le t_i) = [S(t_{j-1}) - S(t_j)]/[1 - S(t_i)]$. Replacing each left-censored observation by these fractional events is the E-step; recomputing KM is the M-step.

Example 8.3: Age at first marijuana use (K&M Section 1.17)

California high-school boys: left censored = used marijuana but cannot recall the age of first use; right censored = never used. Initial KM ignoring the 12 left-censored boys (K&M Table 5.1):

t_i	10	11	12	13	14	15	16	17	18	> 18	Total
c_i	0	0	0	1	2	3	2	3	1	0	12
d_i	4	12	19	24	20	13	3	1	0	4	100
r_i	0	0	2	15	24	18	14	6	0	0	79
Y_i	179	175	163	142	103	59	28	11	4	4	
$S_0(t_i)$	.978	.911	.804	.669	.539	.420	.375	.341	.341	0	

Step 1 at $t_4 = 13$ (one left-censored boy, $1 - S_0(13) = 0.331$):

$$p_{41} = \tfrac{1.000-0.978}{0.331} = 0.067, \quad p_{42} = \tfrac{0.978-0.911}{0.331} = 0.202,$$

$$p_{43} = \tfrac{0.911-0.804}{0.331} = 0.320, \quad p_{44} = \tfrac{0.804-0.669}{0.331} = 0.410.$$

Doing this for every i with $c_i > 0$ (K&M Table 5.2; e.g. the column $i = 9$ reads $0.034, 0.102, 0.161, 0.206, 0.197, 0.180, 0.068, 0.052, 0.000$), Step 2 gives

$$\hat{d}_1 = 4 + 0.067(1) + 0.048(2) + 0.039(3) + 0.036(2) + 0.034(3) + 0.034(1) = 4.487.$$

Step 3 (K&M Table 5.3).

t_i	10	11	12	13	14	15	16	17	18	> 18
$\hat{d}_i$	4.487	13.461	21.313	26.963	22.437	14.714	3.417	1.207	0	4
Y_i	191.000	186.513	173.052	149.739	107.775	61.338	28.624	11.207	4	4
$S_1(t_i)$	.977	.906	.794	.651	.516	.392	.345	.308	.308	0

Iterate until convergence. Note $Y_1 = 191 = 179 + 12$: all left-censored boys are now accounted for as (fractional) events.

3 Turnbull's Iterative Method (1976) for Interval-Censored Data

Let $0 = \tau_0 < \tau_1 < \cdots < \tau_m$ be a grid containing all L_i, R_i. Define $\alpha_{ij} = 1$ if $(\tau_{j-1}, \tau_j] \subseteq (L_i, R_i]$ and 0 otherwise, i.e. whether the event of subject i could have occurred at τ_j.

Definition 8.4: Turnbull (1976) algorithm

Step 1. Probabilities of an event at τ_j: $p_j = S(\tau_{j-1}) - S(\tau_j)$, $j = 1, \ldots, m$.

Step 2. Expected number of events at τ_j: $d_j = \sum_{i=1}^{n} \frac{\alpha_{ij} p_j}{\sum_k \alpha_{ik} p_k}$ (each subject's unit mass is split over its admissible grid points in proportion to p_j).

Step 3. Estimated number at risk: $Y_j = \sum_{k=j}^{m} d_k$.

Step 4. Update the product-limit estimator from these pseudo-data; stop if it is close to the previous S at all τ_j, else repeat Steps 1–3.

Example 8.5: Time to cosmetic deterioration, early breast cancer (radiotherapy only)

46 patients; data are intervals $(L, R]$ in months or right-censored "$\geq$" values: $(0,7]$; $(0,8]$; $(0,5]$; $(4,11]$; $(5,12]$; $(5,11]$; $(6,10]$; $(7,16]$; $(7,14]$; $(11,15]$; $(11,18]$; ≥ 15; ≥ 17; $(17,25]$; $(17,25]$; ≥ 18; $(19,35]$; $(18,26]$; ≥ 22; ≥ 24; ≥ 24; $(25,37]$; $(26,40]$; $(27,34]$; ≥ 32; ≥ 33; ≥ 34; $(36,44]$; $(36,48]$; ≥ 36; ≥ 36; $(37,44]$; ≥ 37; ≥ 37; ≥ 37; ≥ 38; ≥ 40; ≥ 45; ≥ 46 ($\times 7$).

- **Initial estimate**: distribute each subject's mass $1/46$ equally over the grid points in $(L_i, R_i]$; e.g. the subject with $(0,7]$ gives $(1/46)(1/4)$ to each of $\tau = 4,5,6,7$.
- With S_0: $p_1 = 1 - 0.979 = 0.021$, $p_2 = 0.979 - 0.955 = 0.024$, etc.
- At $\tau = 4$, three intervals contain it ($(0,8]$, $(0,7]$, $(0,5]$), so

$$d_1 = \frac{0.021}{0.021+0.024+0.021+0.029+0.031} + \frac{0.021}{0.021+0.024+0.021+0.029} + \frac{0.021}{0.021+0.024} = 0.842.$$

- One iteration changes S only slightly at early times (e.g. $0.979 \to 0.982$ at $\tau = 4$) but more in the tail (e.g. $0.229 \to 0.341$ at $\tau = 46$); iterate to convergence.

4 The Cohort Life Table

4.1 Setting

A **cohort** is a group of individuals with a common origin from which event times are measured. Each event or censoring time is recorded only as falling in one of $k+1$ adjacent, non-overlapping intervals $(a_{j-1}, a_j]$, $j = 1, \ldots, k+1$. A traditional cohort life table presents the actual mortality experience of the cohort from the birth of each individual to the death of the last surviving member; it summarizes time-to-event information over these intervals (grouped data).

Example 8.6: NLSY weaning data

National Labor Survey of Youth (K&M Section 1.14): 927 first-born infants who were breast fed and have complete information. Response: duration of breast feeding in weeks, with an indicator of whether the baby was weaned (event) or lost to follow-up (censored). First rows: weeks 0–2: 927 entering, 2 withdrawn; 2–3: 848 entering, 3 withdrawn; 3–5: 774 entering, 6 withdrawn; ...

4.2 Construction

Definition 8.7: Life-table quantities

1. Intervals $I_j = (a_{j-1}, a_j]$, $j = 1, \ldots, k+1$, $a_0 = 0$, $a_{k+1} = \infty$; each event/censoring time falls in exactly one of them.
2. Y'_j: number entering I_j without having experienced the event.
3. W_j: number lost to follow-up or withdrawn alive in I_j.
4. $Y_j = Y'_j - W_j/2$: estimated number at risk in I_j, assuming censoring times are uniform

over the interval (each withdrawal is at risk for half the interval on average) — the *actuarial* adjustment.

5. d_j: number of events in I_j.

6. Survival at the start of I_j ($\hat{S}(a_0) = 1$):

$$\hat{S}(a_j) = \hat{S}(a_{j-1})[1 - d_j/Y_j] = \prod_{i=1}^{j} [1 - d_i/Y_i]. \tag{8.1}$$

7. Density at the midpoint $a_{mj} = (a_{j-1} + a_j)/2$:

$$\hat{f}(a_{mj}) = \frac{\hat{S}(a_{j-1}) - \hat{S}(a_j)}{a_j - a_{j-1}}. \tag{8.2}$$

8. Hazard at the midpoint (with $\hat{S}(a_{mj})$ taken as the average of the endpoint values):

$$\hat{h}(a_{mj}) = \frac{\hat{f}(a_{mj})}{\hat{S}(a_{mj})} = \frac{2\hat{f}(a_{mj})}{\hat{S}(a_j) + \hat{S}(a_{j-1})} = \frac{d_j}{(a_j - a_{j-1})(Y_j - d_j/2)}. \tag{8.3}$$

Derivation: deriving (8.3)

Write $\hat{q}_j = d_j/Y_j$, $\hat{p}_j = 1 - \hat{q}_j$ and $\Delta_j = a_j - a_{j-1}$. Then $\hat{S}(a_j) = \hat{S}(a_{j-1})\hat{p}_j$ and $\hat{f}(a_{mj}) = \hat{S}(a_{j-1})\hat{q}_j/\Delta_j$, so

$$\hat{h}(a_{mj}) = \frac{2\hat{S}(a_{j-1})\hat{q}_j/\Delta_j}{\hat{S}(a_{j-1})(1 + \hat{p}_j)} = \frac{2\hat{q}_j}{\Delta_j(1 + \hat{p}_j)} = \frac{2d_j/Y_j}{\Delta_j(2 - d_j/Y_j)} = \frac{d_j}{\Delta_j(Y_j - d_j/2)}.$$

Interpretation: events per unit of person-time, where those who fail in I_j are credited with half the interval.

4.3 Standard errors

Theorem 8.8: Estimated standard deviations

$$\widehat{\mathrm{SD}}[\hat{S}(a_{j-1})] = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{d_i}{Y_i(Y_i - d_i)}} = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i}}, \tag{8.4}$$

$$\widehat{\mathrm{SD}}[\hat{f}(a_{mj})] = \frac{\hat{S}(a_{j-1})\hat{q}_j}{a_j - a_{j-1}}\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i} + \frac{\hat{p}_j}{Y_j\hat{q}_j}}, \tag{8.5}$$

$$\widehat{\mathrm{SD}}[\hat{h}(a_{mj})] = \left\{\frac{1 - [\hat{h}(a_{mj})(a_j - a_{j-1})/2]^2}{Y_j\hat{q}_j}\right\}^{1/2} \hat{h}(a_{mj}). \tag{8.6}$$

(8.4) is Greenwood's formula; the SD of $\hat{S}(a_0) = 1$ is 0.

4.4 Other summary quantities

1. **Mean**: $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. If the largest observation is censored, $\hat{S}$ stays flat and the tail must be completed in one of the three ways of Lecture 5 (Efron, Gill, Brown–Hollander–Kowar).

2. **Median** by linear interpolation: find I_j with $\hat{S}(a_{j-1}) \geq 0.5 \geq \hat{S}(a_j)$, then

$$\hat{x}_{0.5} = a_{j-1} + \frac{[\hat{S}(a_{j-1}) - 0.5](a_j - a_{j-1})}{\hat{S}(a_{j-1}) - \hat{S}(a_j)} = a_{j-1} + \frac{\hat{S}(a_{j-1}) - 0.5}{\hat{f}(a_{mj})}. \tag{8.7}$$

3. **Mean residual life** at the start of I_i:

$$\widehat{\mathrm{mr}}(a_{i-1}) = \frac{\int_{a_{i-1}}^\infty \hat{S}(t)\,\mathrm{d}t}{\hat{S}(a_{i-1})} = \frac{\sum_{j=i}^k \hat{S}(a_{j-1})(a_j - a_{j-1})}{\hat{S}(a_{i-1})}.$$

4. **Median residual life**: $\mathrm{mdrl}(a_{i-1})$ solves $S(a_{i-1})/2 = S(a_{i-1} + \mathrm{mdrl}(a_{i-1}))$ — the time increment beyond which half of those alive at a_{i-1} are expected to survive.

Example 8.9: life table for weaning (K&M Table 5.6)

Weeks	Y'_j	W_j	Y_j	d_j	$\hat{S}(a_{j-1})$	$\hat{f}(a_{mj})$	$\hat{h}(a_{mj})$	SD $\hat{S}$	SD $\hat{f}$	SD $\hat{h}$
0–2	927	2	926	77	1.0000	0.0416	0.0434	0	0.0045	0.0049
2–3	848	3	846.5	71	0.9168	0.0769	0.0875	0.0091	0.0088	0.0104
3–5	774	6	771	119	0.8399	0.0648	0.0836	0.0121	0.0055	0.0076
5–7	649	9	644.5	75	0.7103	0.0413	0.0618	0.0149	0.0046	0.0071
7–11	565	7	561.5	109	0.6276	0.0305	0.0537	0.0160	0.0027	0.0051
11–17	449	5	446.5	148	0.5058	0.0279	0.0662	0.0166	0.0021	0.0053
17–25	296	3	294.5	107	0.3381	0.0154	0.0555	0.0158	0.0014	0.0052
25–37	186	0	186	74	0.2153	0.0071	0.0414	0.0138	0.0008	0.0047
37–53	112	0	112	85	0.1296	0.0061	0.0764	0.0114	0.0006	0.0066
53–	27	0	27	27	0.0313	—	—	0.0059	—	—

Check of row 1: $Y_1 = 927 - 2/2 = 926$; $\hat{q}_1 = 77/926 = 0.0832$, so $\hat{S}(2) = 0.9168$; $\hat{f} = (1 - 0.9168)/2 = 0.0416$; $\hat{h} = 77/\{2(926 - 38.5)\} = 0.0434$.

Median: $\hat{S}(11) = 0.5058 \geq 0.5 \geq \hat{S}(17) = 0.3381$, so $\hat{x}_{0.5} = 11 + \dfrac{(0.5058 - 0.5)(17 - 11)}{0.5058 - 0.3381} \approx$ 11.2 weeks.

Hazard (K&M Fig. 5.3): high (about 0.08–0.09 per week) in the first 5 weeks, about 0.05–0.07 through week 25, lowest (about 0.04) in weeks 25–37, then rising again in weeks 37–53.

5 Selected exercises

Motivation. Turnbull's algorithm and the life table were presented as recipes. These exercises show what the algorithm converges to, where the actuarial hazard (8.3) comes from, and how its standard error (8.6) follows from the delta method.

Exercise 8.1 (Turnbull's algorithm and the likelihood)

In the interval-censored setting, let $p = (p_1, \dots, p_m)$ range over the simplex $\Delta = \{p_j \geq 0,\ \sum_j p_j = 1\}$, assume each subject has some $\alpha_{ij} = 1$, and let

$$\ell(p) = \sum_{i=1}^{n} \log\Big(\sum_{j=1}^{m} \alpha_{ij} p_j\Big), \qquad D_j(p) = \frac{\partial \ell}{\partial p_j} = \sum_{i=1}^{n} \frac{\alpha_{ij}}{\sum_k \alpha_{ik} p_k},$$

the log-likelihood (subject i contributes $P(L_i < X \leq R_i)$) and its partial derivatives, at p with $\ell(p) > -\infty$.
(a) Show that one pass of Steps 1–4 maps p to $T(p)$ with $T(p)_j = d_j/n = p_j D_j(p)/n$.
(b) Show that every maximizer of ℓ over Δ is a fixed point of T.
(c) Conversely, show that a fixed point p of T with $D_j(p) \leq n$ for every j with $p_j = 0$ maximizes ℓ over Δ.

Solution [Supplement].

(a)

Step 1. By Step 2 of the algorithm, $d_j = p_j \sum_i \alpha_{ij} / \sum_k \alpha_{ik} p_k = p_j D_j$. Summing over j, each subject contributes $\sum_j \alpha_{ij} p_j / \sum_k \alpha_{ik} p_k = 1$, so $\sum_j d_j = n$; equivalently $\sum_j p_j D_j = n$.

Step 2. The pseudo-data have no censoring: $Y_j = \sum_{k \geq j} d_k$, so $Y_1 = n$ and $Y_{j+1} = Y_j - d_j$ (with $Y_{m+1} = 0$). As in Exercise 5.1(a), the product-limit estimate telescopes to $S_{\text{new}}(\tau_j) = Y_{j+1}/n$ (if some $Y_l = 0$, the previous factor is already 0). Hence $S_{\text{new}}(\tau_{j-1}) - S_{\text{new}}(\tau_j) = (Y_j - Y_{j+1})/n = d_j/n$.

(b)

Step 1. Let p maximize ℓ; then $\ell(p) > -\infty$, so every $\sum_k \alpha_{ik} p_k > 0$. For $q \in \Delta$ and $\varepsilon \in [0, 1)$ these sums stay positive along $p + \varepsilon(q - p)$, and maximality gives the one-sided directional derivative

$$\sum_j D_j(p)(q_j - p_j) = \sum_j D_j q_j - n \leq 0.$$

Step 2. With $q = e_j$ (the j-th vertex): $D_j \leq n$ for all j.

Step 3. If $0 < p_j < 1$, take $q = (p - p_j e_j)/(1 - p_j) \in \Delta$. Then $\sum_k D_k q_k = (n - p_j D_j)/(1 - p_j)$, and Step 1 gives $p_j(n - D_j)/(1 - p_j) \leq 0$, i.e. $D_j \geq n$. If $p_j = 1$, then $D_j = n$ by $\sum_k p_k D_k = n$. So $D_j = n$ whenever $p_j > 0$.

Step 4. By (a), $T(p)_j = p_j D_j / n = p_j$ if $p_j > 0$, and $T(p)_j = 0 = p_j$ if $p_j = 0$.

(c)

Step 1. $T(p) = p$ means $p_j D_j = n p_j$, so $D_j = n$ where $p_j > 0$; with the hypothesis, $D_j \leq n$ for all j.

Step 2. Let $q \in \Delta$ with $\ell(q) > -\infty$ (otherwise there is nothing to prove). The function $g(\varepsilon) = \ell(p + \varepsilon(q - p))$ is concave on $[0, 1]$ (a sum of logs of positive affine functions), so $g(1) \leq g(0) + g'(0+)$:

$$\ell(q) \leq \ell(p) + \sum_j D_j(q_j - p_j) \leq \ell(p) + n \sum_j q_j - n = \ell(p).$$

So Turnbull's iteration is a fixed-point (EM) scheme for the nonparametric MLE; a limit with some $p_j = 0$ must still be checked against $D_j \leq n$. ■

Exercise 8.2 (The actuarial hazard is a piecewise-exponential MLE)

Suppose the hazard is constant on each life-table interval, $h(x) = h_j$ for $x \in I_j$, and right-censored data (T_i, δ_i) are observed exactly.

(a) Show that the MLE is $\hat h_j = d_j/E_j$, where $E_j = \sum_i e_{ij}$ is the total time at risk in I_j, $e_{ij} = \max\{0, \min(T_i, a_j) - a_{j-1}\}$.
(b) If every death and withdrawal in I_j is placed at the midpoint of I_j, show that $\hat h_j$ equals (8.3).

Solution [Supplement].

(a)

Step 1. By the right-censoring likelihood of Lecture 4, subject i contributes $h(T_i)^{\delta_i} e^{-H(T_i)}$, and $H(T_i) = \int_0^{T_i} h = \sum_j h_j e_{ij}$. If $\delta_i = 1$ and $T_i \in I_j$, then $h(T_i) = h_j$.

Step 2. Multiplying over i,

$$L(h_1, \dots, h_{k+1}) = \prod_j h_j^{d_j} e^{-h_j E_j}, \qquad \log L = \sum_j \{d_j \log h_j - h_j E_j\}.$$

Each term is strictly concave in h_j when $d_j > 0$, with derivative $d_j/h_j - E_j$ vanishing at $h_j = d_j/E_j$; when $d_j = 0$ the term $-h_j E_j$ is maximized at $h_j = 0 = d_j/E_j$.

(b) Of the Y_j' subjects entering I_j, $Y_j' - d_j - W_j$ survive the interval (exposure Δ_j each), and the $d_j + W_j$ who die or withdraw contribute $\Delta_j/2$ each. Thus

$$E_j = \Delta_j\Big\{Y_j' - d_j - W_j + \frac{d_j + W_j}{2}\Big\} = \Delta_j\Big(Y_j' - \frac{W_j}{2} - \frac{d_j}{2}\Big) = \Delta_j\Big(Y_j - \frac{d_j}{2}\Big),$$

and $\hat h_j = d_j/\{\Delta_j(Y_j - d_j/2)\}$, which is (8.3). ■

Exercise 8.3 (Standard error of the life-table hazard)

Treat $\hat q_j = d_j/Y_j$ as a binomial proportion, $\mathrm{Var}(\hat q_j) \approx \hat q_j \hat p_j / Y_j$. Derive (8.6) by the delta method.

Solution [Supplement].

Step 1. By the derivation of (8.3), $\hat h = g(\hat q)$ with $g(q) = \dfrac{2q}{\Delta(2-q)}$ (index j dropped), and

$$g'(q) = \frac{2(2-q) + 2q}{\Delta(2-q)^2} = \frac{4}{\Delta(2-q)^2}, \qquad \widehat{\mathrm{SD}}(\hat h) \approx \frac{4}{\Delta(2-\hat q)^2}\Big(\frac{\hat q \hat p}{Y}\Big)^{1/2}.$$

Step 2. Put $u = \hat h \Delta/2 = \hat q/(2-\hat q)$. Then $1 + u = 2/(2-\hat q)$ and $1 - u = 2\hat p/(2-\hat q)$, so $1 - u^2 = 4\hat p/(2-\hat q)^2$.

Step 3. The right-hand side of (8.6) is therefore

$$\hat h\Big(\frac{1-u^2}{Y\hat q}\Big)^{1/2} = \frac{2\hat q}{\Delta(2-\hat q)} \cdot \frac{2}{2-\hat q}\Big(\frac{\hat p}{Y\hat q}\Big)^{1/2} = \frac{4}{\Delta(2-\hat q)^2}\Big(\frac{\hat q \hat p}{Y}\Big)^{1/2},$$

the expression of Step 1. ■

Lecture 9. Hypothesis Testing (Part 1)

1 Overview

1.1 One-sample problems

Rationale. A special pattern in a KM curve may suggest a parametric model; parametric models are useful (efficiency; handling more complex sampling schemes), but once adopted we must check whether the model fits the data adequately. Hypotheses:

$$H_0 : S(t) = S_0(t), \qquad H_0 : H(t) = H_0(t), \qquad H_0 : h(t) = h_0(t) \qquad \text{for all } t \le \tau,$$

with S_0, H_0, h_0 *known*. E.g. under $X \sim \text{Exp}(\lambda)$: $H_0(t) = \lambda t$, $h_0(t) = \lambda$, $S_0(t) = e^{-\lambda t}$.

1.2 Two- and K-sample problems

The study population may be heterogeneous, with groups or key factors relevant to survival. With two groups we compare

$$H_0 : S_1(t) = S_2(t) \iff H_1(t) = H_2(t) \iff h_1(t) = h_2(t), \qquad t \in [0, \tau],$$

equivalent hypotheses with *no* known form. The K-sample case is the obvious extension: $H_0 : S_1(t) = \cdots = S_K(t)$, $t \in [0, \tau]$.

1.3 How to test?

Without censoring: one-sample t test or Kolmogorov–Smirnov; two-/K-sample t tests and ANOVA (approximately normal data) or Wilcoxon rank-sum and Kruskal–Wallis (skewed data). The difficulty is censoring (and truncation); our focus is censoring.

2 One-Sample Test with Right-Censored Data

Data $\{(T_i, \delta_i)\}$, $T_i = \min(X_i, C_i)$, $\delta_i = I(X_i \le C_i)$, X_i with hazard $h(t)$. Test $H_0 : h(t) = h_0(t)$, equivalently $H_0 : H(t) = H_0(t)$, for $t \in [0, \tau]$, where τ is typically the largest observed study time. Notation: $t_1 < \cdots < t_D$ the distinct death times, $d_j = \sum_i I(T_i = t_j, \delta_i = 1)$, $Y_j = Y(t_j) = \sum_i I(T_i \ge t_j)$.

Definition 9.1: One-sample weighted statistic

Let $W(t)$ be a weight function with $W(t) = 0$ whenever $Y(t) = 0$. Compare the Nelson–Aalen increments with the hypothesized hazard:

$$Z(\tau) = O(\tau) - E(\tau) = \sum_{j=1}^{D} W(t_j) \frac{d_j}{Y(t_j)} - \int_0^\tau W(s) h_0(s)\, \mathrm{d}s.$$

Under H_0, its variance is

$$V[Z(\tau)] = \int_0^\tau W^2(s) \frac{h_0(s)}{Y(s)}\, \mathrm{d}s,$$

and for large samples $Z(\tau)^2 / V[Z(\tau)] \sim \chi^2_1$.

Remark 9.2: Intuition

$\tilde{H}(t) = \sum_{t_j \le t} d_j/Y_j$ estimates $H(t)$; the statistic is a weighted distance $\int W \, \mathrm{d}(\tilde{H} - H_0)$. Each increment d_j/Y_j has (Poisson) variance $\approx h_0 \, \mathrm{d}s/Y$, which gives $V[Z]$.

- The χ^2 statistic only serves two-sided alternatives.
- One-sided: for $H_A : h(t) > h_0(t)$ for some $t \le \tau$, reject for large positive $T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$; for $H_A : h(t) < h_0(t)$, reject for large negative T_1.

2.1 One-sample log-rank test

Theorem 9.3: $W(t) = Y(t)$: observed vs. expected

Allow left truncation: T_i = time on study, L_i = delayed entry time. With τ the largest time on study,

$$O(\tau) = \sum_{j=1}^{D} Y(t_j) \frac{d_j}{Y(t_j)} = \sum_{j=1}^{D} \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1) = \sum_{i=1}^{n} I(\delta_i = 1),$$

$$E(\tau) = \int_0^\tau Y(s) h_0(s) \, \mathrm{d}s = \sum_{i=1}^{n} \int_0^\tau I(L_i \le s \le T_i) h_0(s) \, \mathrm{d}s = \sum_{i=1}^{n} [H_0(T_i) - H_0(L_i)].$$

O = observed number of events at or before τ; E = expected number under H_0. Also $V[Z(\tau)] = \int_0^\tau Y(s) h_0(s) \, \mathrm{d}s = E(\tau)$, so the test statistic is $(O - E)^2/E$.

2.2 Harrington–Fleming weights

$$W_{HF}(t) = Y(t) S_0(t)^p [1 - S_0(t)]^q, \qquad p, q \ge 0, \quad S_0(t) = \exp[-H_0(t)].$$

- $p \gg q$: more weight on **early** departures from H_0;
- $q \gg p$: more weight on **late** departures;
- $p = q$: more weight on departures in the **mid-range**;
- $p = q = 0$: the one-sample log-rank test.

Example 9.4: IBCSG Trial 6

International Breast Cancer Study Group: a multi-center trial of adjuvant chemotherapy after surgical removal of the tumour; 735 subjects. The NA cumulative hazard plotted against time is roughly a straight line up to 100 months, suggesting an exponential distribution; eyeballing the slope gives $\hat{\lambda} \approx 0.005$. Formal one-sample log-rank tests ($W = Y$):

H_0	$T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$	p-value
$H_0(t) = 0.0045t$	-0.002	0.998
$H_0(t) = 0.004t$	1.8	0.07
$H_0(t) = 0.005t$	-1.4	0.16
$H_0(t) = 0.006t$	-4.1	< 0.01

$H_0(t) = 0.0045t$ fits best. (Sign check: a larger hypothesized rate inflates E, making $O - E$ negative.)

3 K-Sample Tests

3.1 Hypotheses

$H_0 : h_1(t) = h_2(t) = \cdots = h_K(t) \ \ \forall t \le \tau \qquad$ vs. $\qquad H_A :$ at least one $h_k(t)$ differs for some $t \le \tau$.

Equivalently in terms of H_k or S_k. Here τ is the largest time at which all groups have at least one subject at risk, typically $\tau = \min_k t_{\max,k}$. The alternative is *global*: reject if at least one population differs from the others at some time.

3.2 Data and notation

Right-censored data $\{(T_{ik}, \delta_{ik}),\ k = 1, \ldots, K,\ i = 1, \ldots, n_k\}$. Pool all data and let $0 < t_1 < \cdots < t_D$ be the distinct death times in the pooled sample:

- d_j, d_{jk}: deaths at t_j in the pooled sample / in group k;
- Y_j, Y_{jk}: number at risk at t_j in the pooled sample / in group k.

3.3 Basic idea and the test statistic

Under H_0 the hazards are equal but unknown, so both the pooled and within-group NA estimators estimate the same H:

$$\tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j} \to \int_0^t h(u)\,\mathrm{d}u, \qquad \tilde{H}_k(t) = \sum_{t_j \le t} I(Y_{jk} > 0)\frac{d_{jk}}{Y_{jk}} \to \int_0^t h(u)\,\mathrm{d}u.$$

Build a test from a weighted difference between $\tilde{H}_k$ and $\tilde{H}$.

Definition 9.5: Weighted K-sample statistic

With $W_k(t) \ge 0$ a random weight, zero when $Y_j = 0$ or $Y_{jk} = 0$:

$$Z_k(\tau) = \sum_{j=1}^{D} W_k(t_j)\Big\{\frac{d_{jk}}{Y_{jk}} - \frac{d_j}{Y_j}\Big\}.$$

Typically $W_k(t_j) = Y_{jk}W(t_j)$ with $W(\cdot)$ common to all groups, giving

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\} \quad = \sum_j W(t_j)\,(\text{observed} - \text{expected deaths in group } k).$$

3.4 Distribution under H_0

Theorem 9.6: Asymptotic normality (via martingale theory)

Under H_0, $(Z_1(\tau), \ldots, Z_K(\tau)) \approx \mathrm{MVN}(\mathbf{0}, \Sigma)$, $\Sigma = (\sigma_{kk'})_{K\times K}$, with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k = 1, \ldots, K,$$

$$\hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k \ne k'.$$

Derivation: hypergeometric origin of the variance

At t_j, condition on the margins: Y_j at risk split into $(Y_{j1}, \dots, Y_{jK})$ with d_j deaths. Under H_0 the deaths are a random draw of d_j out of Y_j without replacement, so $(d_{j1}, \dots, d_{jK})$ is multivariate hypergeometric with

$$\mathrm{E}(d_{jk}) = d_j \frac{Y_{jk}}{Y_j}, \quad \mathrm{Var}(d_{jk}) = d_j \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}, \quad \mathrm{Cov}(d_{jk}, d_{jk'}) = -d_j \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}.$$

The factor $(Y_j - d_j)/(Y_j - 1)$ is the finite-population correction (it equals 1 when $d_j = 1$). Summing over (approximately uncorrelated) death times gives $\hat{\sigma}_{kk'}$.

3.5 Linear dependency and the final statistic

Theorem 9.7: $\sum_k Z_k(\tau) = 0$

$$\sum_{k=1}^{K} Z_k(\tau) = \sum_{j=1}^{D} W(t_j) \sum_{k=1}^{K} \Big\{ d_{jk} - d_j \frac{Y_{jk}}{Y_j} \Big\} = \sum_{j=1}^{D} W(t_j)(d_j - d_j) = 0,$$

since $\sum_k d_{jk} = d_j$ and $\sum_k Y_{jk} = Y_j$. Hence $Z_1, \dots, Z_K$ are linearly dependent and Σ is singular.

Eliminate the redundancy by dropping one (arbitrary) $Z_K(\tau)$:

$$(Z_1, \dots, Z_{K-1}) \approx \mathrm{MVN}(\mathbf{0}, \Sigma_{(K-1)\times(K-1)}),$$

and

$$\chi^2 = (Z_1(\tau), \dots, Z_{K-1}(\tau))\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, (Z_1(\tau), \dots, Z_{K-1}(\tau))^T \overset{H_0}{\sim} \chi^2_{K-1}.$$

Reject at level α if $\chi^2 > \chi^2(K-1, 1-\alpha)$. (The value does not depend on which Z_k is dropped.)

3.6 Special case $K = 2$

$$Z = \frac{Z_1(\tau)}{\sqrt{\hat{V}(Z_1(\tau))}} = \frac{\sum_{j=1}^{D} W(t_j)(d_{j1} - Y_{j1}d_j/Y_j)}{\sqrt{\sum_{j=1}^{D} W^2(t_j)\frac{Y_{j1}}{Y_j}(1 - \frac{Y_{j1}}{Y_j})\frac{Y_j - d_j}{Y_j - 1}d_j}} \overset{H_0}{\approx} N(0,1).$$

For $K = 2$, a two-sided level-α test rejects if $|Z| > Z_{1-\alpha/2}$; one-sided tests are possible. For $K > 2$ we must use the χ^2 test, which only handles two-sided alternatives.

4 Choice of the Weight Function

Different weights change the power against specific alternatives; they are extremely well studied. All have the form $W_k(t_j) = Y_{jk}W(t_j)$.

1. **Log-rank**: $W(t) = 1$. Optimal power against **proportional hazards** alternatives $h_1(t) = c_2 h_2(t) = \cdots = c_K h_K(t)$.
2. **Gehan** (generalized Wilcoxon): $W(t) = Y(t) = \sum_k \sum_i I(T_{ik} \geq t)$, so $Z_k(\tau) = \sum_j (Y_j d_{jk} - d_j Y_{jk})$. Commonly used.
3. **Tarone–Ware**: $W(t) = f(Y(t))$ with known increasing f; $f(u) = u$ is Gehan, the original Tarone–Ware uses $f(u) = u^{1/2}$, also $f(u) = \log u$. Gives relatively larger weight to death times with larger risk sets.

4. **Peto–Peto**: $W(t) = \tilde{S}(t)$, $\tilde{S}(t) = \prod_{t_j \le t}\left(1 - \frac{d_j}{Y_j+1}\right)$, a KM-like estimator from the pooled sample.
5. **Modified Peto–Peto** (Andersen, 1982): $W(t) = \tilde{S}(t)Y(t)/(Y(t)+1)$. Both Peto–Peto tests depend on the pooled survival function.
6. **Fleming–Harrington**: $W(t_j) = \hat{S}(t_{j-1})^p\left[1 - \hat{S}(t_{j-1})\right]^q$, $p, q \ge 0$, $\hat{S}$ the pooled KM.
   - $p = q = 0$: log-rank;
   - $p > 0, q = 0$: larger weights early ⇒ sensitive to early differences among hazards;
   - $p = 0, q > 0$: larger weights late ⇒ sensitive to late differences.

Remark 9.8: Why the left-continuous $\hat{S}(t_{j-1})$

The weight at t_j must be *predictable* — known just before t_j — so it uses survival information up to t_{j-1}, not the deaths at t_j themselves. This is what makes Z_k a martingale-based statistic.

5 Catheter Placement in Kidney Dialysis Patients

Clinical trial (K&M Section 1.4) comparing time to cutaneous exit-site infection for catheters placed surgically (group 1) vs. percutaneously (group 2). From the KM curves (K&M Fig. 7.1), the surgical group does better early, but the percutaneous group has higher infection-free probability later (about 0.8 vs. 0.2 near month 27). The curves *cross*, so the choice of weight matters:

- weighting early times heavily (e.g. Gehan) may suggest the surgical group is better;
- weighting late times heavily may suggest the percutaneous group is better.

Relative weights (K&M Fig. 7.2): Gehan and Tarone–Ware decrease steeply with time; log-rank is flat; Fleming–Harrington with $q > 0$ increase with time.

Example 9.9: Log-rank computation (K&M Table 7.2, selected rows)

t_j	Y_{j1}	d_{j1}	Y_{j2}	d_{j2}	Y_j	d_j	$Y_{j1}\frac{d_j}{Y_j}$	$d_{j1} - Y_{j1}\frac{d_j}{Y_j}$	$\frac{Y_{j1}}{Y_j}\left(1-\frac{Y_{j1}}{Y_j}\right)\frac{Y_j-d_j}{Y_j-1}d_j$
0.5	43	0	76	6	119	6	2.168	−2.168	1.326
1.5	43	1	60	0	103	1	0.417	0.583	0.243
2.5	42	0	56	2	98	2	0.857	−0.857	0.485
4.5	36	2	43	0	79	2	0.911	1.089	0.490
							⋮		
26.5	2	1	3	0	5	1	0.400	0.600	0.240
SUM		15		11		26	11.036	3.964	6.211

So $Z_1(\tau) = 15 - 11.036 = 3.964$, $\hat{\sigma}_{11} = 6.211$, $\chi^2 = 3.964^2/6.211 = 2.53$, $p = 0.112$: no significant difference by the log-rank test.

Example 9.10: Comparison of two-sample tests (K&M Table 7.3)

Test	$W(t_j)$	$Z_1(\tau)$	σ^2_{11}	χ^2	p-value
Log-rank	1	3.96	6.21	2.53	0.112
Gehan	Y_j	−9	38862	0.002	0.964
Tarone–Ware	$Y_j^{1/2}$	13.20	432.83	0.40	0.526
Peto–Peto	$\tilde{S}(t_j)$	2.47	4.36	1.40	0.237
Modified Peto–Peto	$\tilde{S}(t_j)Y_j/(Y_j+1)$	2.31	4.20	1.28	0.259
FH $p=0, q=1$	$1-\hat{S}(t_{j-1})$	1.41	0.21	9.67	0.002
FH $p=1, q=0$	$\hat{S}(t_{j-1})$	2.55	4.69	1.39	0.239
FH $p=1, q=1$	$\hat{S}(t_{j-1})[1-\hat{S}(t_{j-1})]$	1.02	0.11	9.83	0.002
FH $p=0.5, q=0.5$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^{0.5}$	2.47	0.66	9.28	0.002
FH $p=0.5, q=2$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^{2}$	0.32	0.01	8.18	0.004

Remark 9.11: Comments

- Gehan's weight puts very heavy weight on early comparisons, leading to a *negative* $Z_1(\tau)$.
- Fleming–Harrington tests with $q > 0$ weight late comparisons and are significant, because the two curves diverge for large t.
- In practice, the choice of weight is an important decision and should be made *before* looking at the data.
- The log-rank test ($W = 1$) is the most popular, followed by Gehan (generalized Wilcoxon rank-sum); both are in most statistical packages.

6 Three-Group Comparison (Bone Marrow Transplant)

Disease-free survival of leukaemia patients after bone marrow transplantation: Group 1 = 38 ALL, Group 2 = 54 AML low-risk, Group 3 = 45 AML high-risk. Test $H_0 : S_1(t) = S_2(t) = S_3(t)$, $t \le \tau = 2204$ days.

Example 9.12: Log-rank test with $K = 3$

Using the first two statistics, $Z_1(2204) = 2.148$, $Z_2(2204) = -14.966$, with

$$\hat{\Sigma} = \begin{pmatrix} 15.9552 & -10.3451 \\ -10.3451 & 20.3398 \end{pmatrix}, \qquad \chi^2 = (2.148,\ -14.966)\,\hat{\Sigma}^{-1} \begin{pmatrix} 2.148 \\ -14.966 \end{pmatrix} = 13.8037.$$

On $K - 1 = 2$ df, $p \approx 0.001$.

Derivation: evaluating the quadratic form [Supplement]

$\det \hat{\Sigma} = 15.9552(20.3398) - 10.3451^2 = 324.53 - 107.02 = 217.51$, and $\hat{\Sigma}^{-1} = \frac{1}{217.51}\begin{pmatrix} 20.3398 & 10.3451 \\ 10.3451 & 15.9552 \end{pmatrix}$. Then

$$\chi^2 = \frac{1}{217.51}\Big[20.3398(2.148)^2 + 2(10.3451)(2.148)(-14.966) + 15.9552(-14.966)^2\Big]$$

$$= \frac{93.85 - 665.14 + 3573.73}{217.51} \approx 13.80.$$

Other weights: Gehan $\chi^2 = 16.24$ ($p = 0.0003$); Tarone–Ware 15.65 ($p = 0.0004$); FH $(1,0)$ 15.67 ($p = 0.0004$); FH $(0,1)$ 6.11 ($p = 0.047$); FH $(1,1)$ 9.93 ($p = 0.007$). The results are robust: all tests suggest that the disease-free survival curves differ among the three groups.

7 Selected exercises

Motivation. Three facts used above without proof: the hypergeometric moments behind $\hat{\Sigma}$, the identity that makes the one-sample log-rank test a score test, and the claim that the K-sample χ^2 does not depend on which Z_k is dropped.

Exercise 9.1 (Hypergeometric moments)

At a death time, $Y \geq 2$ subjects are at risk, Y_k of them in group k ($\sum_k Y_k = Y$), and the d deaths form a uniformly random subset of size d of the Y subjects. Let d_k be the number of deaths in group k. Prove the formulas for $\mathrm{E}(d_k)$, $\mathrm{Var}(d_k)$ and $\mathrm{Cov}(d_k, d_{k'})$ stated in this lecture.

Solution [Supplement].

Step 1. Label the subjects $1, \dots, Y$ and let I_i indicate that subject i dies. By symmetry $P(I_i = 1) = d/Y$, and for $i \neq l$, $P(I_i = I_l = 1) = \binom{Y-2}{d-2}/\binom{Y}{d} = \dfrac{d(d-1)}{Y(Y-1)}$.

Step 2. Hence $\mathrm{Var}(I_i) = \frac{d}{Y}(1 - \frac{d}{Y})$ and, for $i \neq l$,

$$\mathrm{Cov}(I_i, I_l) = \frac{d(d-1)}{Y(Y-1)} - \frac{d^2}{Y^2} = \frac{d\{(d-1)Y - d(Y-1)\}}{Y^2(Y-1)} = -\frac{d(Y-d)}{Y^2(Y-1)} =: -c.$$

Step 3. With $d_k = \sum_{i \in \text{group } k} I_i$, $\mathrm{E}(d_k) = dY_k/Y$ and

$$\mathrm{Var}(d_k) = Y_k \frac{d(Y-d)}{Y^2} - Y_k(Y_k - 1)c = \frac{Y_k d(Y-d)}{Y^2}\Big\{1 - \frac{Y_k - 1}{Y - 1}\Big\} = d\,\frac{Y_k}{Y}\Big(1 - \frac{Y_k}{Y}\Big)\frac{Y-d}{Y-1}.$$

Step 4. For $k \neq k'$ all $Y_k Y_{k'}$ pairs are distinct, so

$$\mathrm{Cov}(d_k, d_{k'}) = -Y_k Y_{k'} c = -d\,\frac{Y_k Y_{k'}}{Y^2}\,\frac{Y-d}{Y-1}.$$

■

Exercise 9.2 (The one-sample log-rank test is a score test)

Let $T = \min(X, C)$, $\delta = I(X \leq C)$ with $X \perp\!\!\!\perp C$, X continuous with cumulative hazard H.
(a) Show that $\mathrm{E}(\delta) = \mathrm{E}\{H(T)\}$.
(b) In the exponential model $h \equiv \lambda$ (no truncation), show that the score statistic for $H_0 : \lambda = \lambda_0$, with the information estimated by $\sum_i T_i/\lambda_0$, equals the one-sample log-rank statistic $(O - E)^2/E$.

Solution [Supplement].

(a) Let $G(s) = P(C \geq s)$. Writing $H(T) = \int_0^\infty I(T \geq s)h(s)\,\mathrm{d}s$ and using Tonelli and independence,

$$\mathrm{E}\{H(T)\} = \int_0^\infty P(X \geq s)G(s)h(s)\,\mathrm{d}s = \int_0^\infty f(s)G(s)\,\mathrm{d}s = P(X \leq C) = \mathrm{E}(\delta),$$

since $P(X \geq s)h(s) = f(s)$ and $P(X \leq C) = \int f(s)P(C \geq s)\,\mathrm{d}s$. (This is $\mathrm{E}\{N(\infty) - \int Y\,\mathrm{d}H\} = 0$ for one subject: the counting process minus its compensator has mean zero.)

(b)

Step 1. With $O = \sum_i \delta_i$ and $\Sigma T = \sum_i T_i$, the log-likelihood is $O \log \lambda - \lambda\,\Sigma T$, with score $U(\lambda) = O/\lambda - \Sigma T$ and Fisher information $\mathcal{I}(\lambda) = \mathrm{E}(O)/\lambda^2$.

Step 2. By (a) under H_0, $\mathrm{E}(O) = \mathrm{E}\{\sum_i H_0(T_i)\} = \lambda_0 \mathrm{E}(\Sigma T)$, so $\mathcal{I}(\lambda_0) = \mathrm{E}(\Sigma T)/\lambda_0$, estimated by $\Sigma T/\lambda_0$.

Step 3. Here $E = \sum_i H_0(T_i) = \lambda_0 \Sigma T$ (Theorem on the one-sample log-rank test with $L_i = 0$). Hence

$$\frac{U(\lambda_0)^2}{\Sigma T/\lambda_0} = \frac{(O - \lambda_0 \Sigma T)^2/\lambda_0^2}{\Sigma T/\lambda_0} = \frac{(O-E)^2}{\lambda_0 \Sigma T} = \frac{(O-E)^2}{E}.$$

■

Exercise 9.3 (The χ^2 does not depend on the dropped group)

Let $\mathbf{Z} = (Z_1, \ldots, Z_K)'$ and $\hat{\Sigma}$ be as in this lecture. For each k let $\mathbf{Z}_{(-k)}$ and $\hat{\Sigma}_{(-k)}$ be obtained by deleting the k-th entry, respectively the k-th row and column, and assume $\hat{\Sigma}_{(-K)}$ is invertible.
(a) Show that $\hat{\Sigma}\mathbf{1} = \mathbf{0}$.
(b) Show that $\hat{\Sigma}_{(-k)}$ is invertible and $\mathbf{Z}'_{(-k)}\hat{\Sigma}^{-1}_{(-k)}\mathbf{Z}_{(-k)} = \mathbf{Z}'_{(-K)}\hat{\Sigma}^{-1}_{(-K)}\mathbf{Z}_{(-K)}$ for every k.

Solution [Supplement].

(a) Fix a death time and write $\pi_k = Y_{jk}/Y_j$, $w = W(t_j)^2 d_j (Y_j - d_j)/(Y_j - 1)$. Its contribution to $\sum_{k'} \hat{\sigma}_{kk'}$ is

$$w\Big\{\pi_k(1-\pi_k) - \sum_{k' \neq k} \pi_k \pi_{k'}\Big\} = w\,\pi_k\Big\{1 - \sum_{k'} \pi_{k'}\Big\} = 0.$$

(b)

Step 1. Relabeling the groups, it suffices to compare $k = 1$ with K. Let P_k be the $(K-1) \times K$ matrix deleting coordinate k, so $\mathbf{Z}_{(-k)} = P_k \mathbf{Z}$ and $\hat{\Sigma}_{(-k)} = P_k \hat{\Sigma} P_k'$.

Step 2. On $V = \{\mathbf{z} : \mathbf{1}'\mathbf{z} = 0\}$ we have $z_K = -\sum_{k<K} z_k$, so $P_1 \mathbf{z} = M P_K \mathbf{z}$ for the $(K-1) \times (K-1)$ matrix M that maps $(z_1, \ldots, z_{K-1})$ to $(z_2, \ldots, z_{K-1}, -\sum_{k<K} z_k)$. M is invertible: from its output one recovers $z_2, \ldots, z_{K-1}$ and then z_1.

Step 3. $\mathbf{Z} \in V$ (Theorem $\sum_k Z_k(\tau) = 0$). By (a) and symmetry, $\mathbf{1}'\hat{\Sigma} = \mathbf{0}'$, so every column of $\hat{\Sigma}$ lies in V. Applying Step 2 to the columns of $\hat{\Sigma}P_1'$ and then, after transposing, to the columns of $\hat{\Sigma}P_K'$,

$$P_1 \hat{\Sigma} P_1' = M P_K \hat{\Sigma} P_1' = M\big(P_1 \hat{\Sigma} P_K'\big)' = M\big(M P_K \hat{\Sigma} P_K'\big)' = M \hat{\Sigma}_{(-K)} M'.$$

Step 4. Hence $\hat{\Sigma}_{(-1)} = M \hat{\Sigma}_{(-K)} M'$ is invertible, and with $\mathbf{Z}_{(-1)} = M \mathbf{Z}_{(-K)}$,

$$\mathbf{Z}'_{(-1)}\hat{\Sigma}^{-1}_{(-1)}\mathbf{Z}_{(-1)} = \mathbf{Z}'_{(-K)} M' (M')^{-1} \hat{\Sigma}^{-1}_{(-K)} M^{-1} M \mathbf{Z}_{(-K)} = \mathbf{Z}'_{(-K)}\hat{\Sigma}^{-1}_{(-K)}\mathbf{Z}_{(-K)}.$$

■

Lecture 10. Hypothesis Testing (Part 2)

1 Review: The K-Sample Test

Right-censored data $\{(T_{ik}, \delta_{ik})\}$; pooled death times $t_1 < \cdots < t_D$; d_j, d_{jk} deaths and Y_j, Y_{jk} at risk at t_j (pooled / group k). With a common weight W,

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\}, \qquad \chi^2 = (Z_1, \ldots, Z_{K-1})\,\hat{\Sigma}_{K-1}^{-1}\,(Z_1, \ldots, Z_{K-1})^T \overset{H_0}{\approx} \chi^2_{K-1},$$

with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}d_j, \qquad \hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

This omnibus test is **not sensitive to a trend alternative**.

2 Tests for Trend

2.1 Order-specific alternatives

$$H_0 : h_1(t) = \cdots = h_K(t) \;\; \forall t \le \tau \qquad \text{vs.} \qquad H_A : h_1(t) \le h_2(t) \le \cdots \le h_K(t) \;\; \forall t \le \tau,$$

with at least one strict inequality; equivalently $H_A : S_1(t) \ge S_2(t) \ge \cdots \ge S_K(t)$.

2.2 Trend test statistic

Objective: increase power by a statistic sensitive to the stochastic ordering.

Definition 10.1: Trend test

For scores $a_1 < a_2 < \cdots < a_K$,

$$Z = \frac{\sum_{k=1}^{K} a_k Z_k(\tau)}{\Big(\sum_{k=1}^{K}\sum_{k'=1}^{K} a_k a_{k'} \hat{\sigma}_{kk'}\Big)^{1/2}} = \frac{\mathbf{a}'\mathbf{Z}}{\sqrt{\mathbf{a}'\hat{\Sigma}\mathbf{a}}},$$

with $\hat{\sigma}_{kk'}$ from the full $K \times K$ $\hat{\Sigma}$. Under H_0 each $Z_k(\tau)$ is a mean-zero martingale, so the linear combination is approximately normal with mean 0, and $Z \approx N(0,1)$. Reject H_0 in favor of H_A at level α if Z exceeds the upper α quantile of $N(0,1)$ (one-sided).

- **Choice of a_k**: heuristically, under H_A, $\mathrm{E}\{Z_1(\tau)\} \le \mathrm{E}\{Z_2(\tau)\} \le \cdots \le \mathrm{E}\{Z_K(\tau)\}$ (higher hazard $\Rightarrow$ more observed than expected deaths). Taking $a_k = k$ (standard) gives large weight to the groups with large expected values under H_A. Other choices may have better properties.
- The full $\hat{\Sigma}$ is singular, but $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ as long as $\mathbf{a}$ is not constant, so no group needs to be dropped.

2.3 Invariance

Theorem 10.2: Invariance under linear transformations of the scores

Let $Z_{\mathbf{a}} = \mathbf{a}'\mathbf{Z}/\sqrt{V[\mathbf{a}'\mathbf{Z}]}$ and $\mathbf{b} = c\mathbf{a} + d\mathbf{1}_K$ with constants $c > 0$, d. Then $Z_{\mathbf{b}}(\tau) = Z_{\mathbf{a}}(\tau)$.

Proof [Supplement]. Since $\sum_k Z_k(\tau) = 0$, i.e. $\mathbf{1}'\mathbf{Z} = 0$ (and hence $\hat{\Sigma}\mathbf{1} = \mathbf{0}$),

$$\mathbf{b}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z} + d\,\mathbf{1}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z}, \qquad V[\mathbf{b}'\mathbf{Z}] = V[c\,\mathbf{a}'\mathbf{Z}] = c^2 V[\mathbf{a}'\mathbf{Z}].$$

So $Z_{\mathbf{b}} = c\,\mathbf{a}'\mathbf{Z}/(c\sqrt{V[\mathbf{a}'\mathbf{Z}]}) = Z_{\mathbf{a}}$. (If $c < 0$ the sign flips.) ■

E.g. scores $(1, 2, 3, 4)$ and $(0, 1, 2, 3)$ or $(-3, -1, 1, 3)$ give the same test.

Example 10.3: larynx cancer by stage

90 larynx cancer patients in stages I–IV. KM curves (K&M Fig. 7.3) are ordered: stage I best, stage IV worst. Give the most severe disease the heaviest weight: $a_k = k$, $k = 1, \ldots, 4$. With log-rank weights,

$$\mathbf{Z}(10.7) = (-7.5660,\ -3.0117,\ 2.9155,\ 7.6623),$$

$$\hat{\Sigma} = \begin{pmatrix} 12.0740 & -4.4516 & -6.2465 & -1.3759 \\ -4.4516 & 7.8730 & -2.7599 & -0.6614 \\ -6.2465 & -2.7599 & 9.9302 & -0.9238 \\ -1.3759 & -0.6614 & -0.9238 & 2.9612 \end{pmatrix}.$$

Then $\mathbf{a}'\mathbf{Z} = -7.566 - 6.023 + 8.747 + 30.649 = 25.806$ and $\mathbf{a}'\hat{\Sigma}\mathbf{a} = 48.152$, so

$$Z = \frac{25.806}{\sqrt{48.152}} = 3.72, \qquad Z^2 = 13.83, \qquad p < 0.0001.$$

(The value 13.83 reported in K&M is Z^2.) With other weights: Tarone–Ware $Z = 4.06$, Gehan 4.22, Peto–Peto 4.13. All have $p < 0.0001$, strong evidence that the hazard rates are in the expected order.

Remark 10.4

Use a trend test only with strong prior information about the group ordering; otherwise it can lose power relative to the omnibus test.

3 Stratified Tests

3.1 Rationale

When comparing K groups we often need to adjust for other covariates. E.g. compare 3 vs. 6 months of chemotherapy in IBCSG adjusting for tumour stage (I–IV): the interest is the treatment, not the stage. One option is a regression model for covariate effects, but model-based adjustment is sensitive to misspecification. A conservative alternative is **stratification**.

3.2 Set-up

Definition 10.5: Stratified K-sample test

$H_0 : h_{1s}(t) = h_{2s}(t) = \cdots = h_{Ks}(t)$, $t \in [0, \tau]$, for each stratum $s = 1, \ldots, M$.

1. Using only the data in stratum s, compute $Z_{ks}(\tau)$, $k = 1, \ldots, K$, and $\hat{\Sigma}_s = \widehat{\mathrm{Cov}}[Z_{1s}(\tau), \ldots, Z_{Ks}(\tau)]$.
2. Global test: $Z_{k\cdot}(\tau) = \sum_{s=1}^{M} Z_{ks}(\tau)$ and $\hat{\Sigma}_{\cdot} = \widehat{\mathrm{Cov}}[Z_{1\cdot}, \ldots, Z_{K\cdot}] = \sum_{s=1}^{M} \hat{\Sigma}_s$ (strata are independent).
3. $\chi^2 = [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]^T \overset{H_0}{\sim} \chi^2_{K-1}$.

Within each stratum, observed and expected deaths are compared only among subjects with the same stratum, so differences in the baseline hazard across strata cannot masquerade as group effects.

Example 10.6: bone marrow transplant, stratified by MTX

The three disease groups (ALL, AML low risk, AML high risk) were compared in Lecture 9 and all tests rejected H_0. A more careful comparison accounts for whether patients received the graft-versus-host prophylactic methotrexate (MTX) or not (NOMTX). Stratified log-rank ($W = 1$):

$$\text{MTX:} \quad (Z_1, Z_2, Z_3) = (1.2816,\ -2.4929,\ 1.2113),$$

$$\hat{\Sigma}_{MTX} = \begin{pmatrix} 6.23336 & -3.42915 & -2.80422 \\ -3.42915 & 5.60732 & -2.17818 \\ -2.80422 & -2.17818 & 4.98239 \end{pmatrix},$$

$$\text{NOMTX:} \quad (Z_1, Z_2, Z_3) = (-0.504,\ -11.235,\ 11.739),$$

$$\hat{\Sigma}_{NOMTX} = \begin{pmatrix} 9.6771 & -6.5535 & -3.1236 \\ -6.5535 & 13.9816 & -7.4281 \\ -3.1236 & -7.4281 & 10.5517 \end{pmatrix}.$$

Summing: $Z_{1\cdot} = 0.7776$, $Z_{2\cdot} = -13.7279$, $Z_{3\cdot} = 12.9503$ (again linearly dependent), and

$$\hat{\Sigma}_{\cdot} = \begin{pmatrix} 15.91046 & -9.98265 & -5.92782 \\ -9.98265 & 19.58892 & -9.60628 \\ -5.92782 & -9.60628 & 15.53409 \end{pmatrix}.$$

Using the first two:

$$\chi^2 = (0.7776,\ -13.7279) \begin{pmatrix} 15.9105 & -9.9827 \\ -9.9827 & 19.5889 \end{pmatrix}^{-1} \begin{pmatrix} 0.7776 \\ -13.7279 \end{pmatrix} = 13.19 \quad (2 \text{ df},\ p \approx 0.001).$$

Remark 10.7: Remarks

Stratified tests have good power against alternatives in the *same direction* in every stratum. Otherwise, Z from one stratum can be cancelled by Z from another, the overall statistic may be near 0, and the test has very low power. In that case, run separate K-sample tests in each stratum.

4 Matched Pairs

Another use of stratification: each matched pair is a stratum. Paired times (T_{1i}, T_{2i}) and indicators $(\delta_{1i}, \delta_{2i})$, $i = 1, \dots, M$; test $H_0 : h_{1i}(t) = h_{2i}(t)$. With $K = 2$, in each pair

$$Z_{1i}(\tau) = \sum_j W(t_j)\Big\{d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big\}, \qquad \sigma_{11i} = \sum_j W(t_j)^2 \frac{Y_{j1}}{Y_j}\Big(1 - \frac{Y_{j1}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}d_j.$$

Derivation: within a single pair

Before the first time in the pair both are at risk: $Y_j = 2$, $Y_{j1} = 1$, so $Y_{j1}/Y_j = 1/2$.

1. $T_{1i} = T_{2i}$: $d_j = \delta_{1i} + \delta_{2i}$,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i} + \delta_{2i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i} - \delta_{2i}}{2},$$

$$\sigma_{11i} = W^2(T_{1i})\tfrac{1}{2}(1 - \tfrac{1}{2})\frac{2 - \delta_{1i} - \delta_{2i}}{2 - 1}(\delta_{1i} + \delta_{2i})$$

$$= \frac{W^2(T_{1i})}{4}(2 - \delta_{1i} - \delta_{2i})(\delta_{1i} + \delta_{2i}).$$

2. $T_{1i} < T_{2i}$: only the first time can be an event with both at risk,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i}}{2}, \qquad \sigma_{11i} = \frac{W^2(T_{1i})}{4}(2 - \delta_{1i})\delta_{1i}.$$

3. $T_{1i} > T_{2i}$: $Z_{1i} = -W(T_{2i})\dfrac{\delta_{2i}}{2}$, $\quad \sigma_{11i} = \dfrac{W^2(T_{2i})}{4}(2 - \delta_{2i})\delta_{2i}$.

(After the first time, one of the pair is gone, so $Y_{j1}/Y_j \in \{0, 1\}$ and later terms contribute nothing.) For 0/1 indicators $(2 - \delta)\delta = \delta$, so:

$$Z_{1i} = \begin{cases} W(T_{1i})/2, & T_{1i} < T_{2i}, \delta_{1i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{1i} = 1, \delta_{2i} = 0, \\ -W(T_{2i})/2, & T_{2i} < T_{1i}, \delta_{2i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{2i} = 1, \delta_{1i} = 0, \\ 0, & \text{otherwise;} \end{cases}$$

$$\hat{\sigma}_{11i} = \begin{cases} W(T_{1i})^2/4, & \text{first case,} \\ W(T_{2i})^2/4, & \text{second case,} \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 10.8: Paired log-rank (sign-type) test

If W does not depend on time (e.g. log-rank, $W \equiv 1$), with $D_1 = \#$ pairs in which member 1 has the event first and $D_2 = \#$ pairs in which member 2 does,

$$Z_{1\cdot} = \frac{W}{2}(D_1 - D_2), \qquad \hat{\sigma}_{11\cdot} = \sum_{i=1}^{M}\hat{\sigma}_{11i} = \frac{W^2}{4}(D_1 + D_2), \qquad \frac{Z_{1\cdot}}{\sqrt{\hat{\sigma}_{11\cdot}}} = \frac{D_1 - D_2}{\sqrt{D_1 + D_2}} \overset{H_0}{\approx} N(0, 1).$$

This is the sign test (McNemar-type): under H_0 each informative pair is equally likely to go either way.

Example 10.9: 6-MP vs. placebo matched pairs

$D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$, so $Z = (18-3)/\sqrt{18+3} = 3.27$ and $p = 2P[Z > 3.27] = 0.001$: sufficient evidence that the relapse rates differ.

5 Renyi-Type Tests

5.1 Rationale

If the hazards of the two groups **cross**, the usual two-sample tests may have low power: early differences cancel late differences of opposite sign. Renyi-type tests are built on the *absolute values of cumulative differences*; a Renyi statistic is the analogue for censored data of the Kolmogorov–Smirnov statistic.

5.2 Set-up and statistic

Test $H_0 : h_1(t) = h_2(t)$, $t \le \tau$, vs. $H_A : h_1(t) \neq h_2(t)$ for some $t \le \tau$. Two samples of sizes n_1, n_2; pooled distinct failure times $t_1 < \cdots < t_D$; d_{j1}, d_{j2} and Y_{j1}, Y_{j2} with $d_j = d_{j1} + d_{j2}$, $Y_j = Y_{j1} + Y_{j2}$.

Definition 10.10: Renyi statistic

Cumulative weighted difference up to t_i:

$$Z(t_i) = \sum_{t_j \le t_i} W(t_j)\Big[d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big], \qquad i = 1, \ldots, D,$$

with standard error at τ (largest t_j with $Y_{j1}, Y_{j2} > 0$)

$$\sigma^2(\tau) = \sum_{t_j \le \tau} W^2(t_j)\frac{Y_{j1}}{Y_j}\frac{Y_{j2}}{Y_j}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

The two-sided test statistic is

$$Q = \frac{\sup\{|Z(t)|,\ t \le \tau\}}{\sigma(\tau)}.$$

Theorem 10.11: Null distribution

Under H_0, Q is approximately distributed as $\sup\{|B(x)|,\ 0 \le x \le 1\}$, B a standard Brownian motion. Critical values are in K&M Table C.5.

Remark 10.12: Why Brownian motion

$Z(t)$ is a martingale in t with variance process $\sigma^2(t)$; rescaling time by $x = \sigma^2(t)/\sigma^2(\tau) \in [0, 1]$, $Z/\sigma(\tau)$ behaves like $B(x)$. The supremum of $|Z|$ catches a large excursion even if $Z(\tau)$ returns near 0.

Comparison with the usual test. The usual statistic is $Z(\tau)/\sqrt{\text{Var}[Z(\tau)]}$ (or its square). When hazards cross, early positive differences are cancelled by later negative ones, and the log-rank test may lack power; the supremum test based on $|Z|$ has greater power for such differences.

Example 10.13: gastric cancer: chemotherapy vs. chemotherapy + radiotherapy

The KM curves cross (K&M Fig. 7.5): chemo + radiation is worse early but better later. Using the log-rank version of the Renyi statistic, $|Z(t_j)|$ (K&M Fig. 7.4) peaks at $t_j = 315$ with value 9.8; $\sigma(\tau = 2363) = 4.46$, so

$$Q = 9.8/4.46 = 2.2, \qquad 0.05 < p < 0.06 \text{ (K\&M Table C.5)},$$

close to rejecting H_0. The usual log-rank test gives $Z(2363) = -2.15$, statistic $-2.15/4.46 = -0.482$, $p = 2P(Z < -0.482) = 0.63$: it misses the crossing difference entirely.

6 Selected exercises

Motivation. Two claims of this lecture deserve a proof: when the trend-test variance $\mathbf{a}'\hat{\Sigma}\mathbf{a}$ is really positive, and why the paired log-rank test is an exact sign test under the null hypothesis.

Exercise 10.1 (When is the trend test defined?)

Write $\pi_{jk} = Y_{jk}/Y_j$ and $w_j = W(t_j)^2 d_j(Y_j - d_j)/(Y_j - 1)$, with $w_j = 0$ if $Y_j = 1$.

(a) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} = \sum_j w_j \Big\{ \sum_k \pi_{jk} a_k^2 - \Big(\sum_k \pi_{jk} a_k \Big)^2 \Big\} \geq 0$.

(b) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ if and only if at some death time with $w_j > 0$ the scores a_k are not all equal over the groups with $Y_{jk} > 0$. In particular, for $a_1 < \cdots < a_K$ it suffices that at one death time with $W(t_j) \neq 0$ and $d_j < Y_j$ at least two groups are at risk.

(c) For $K = 2$ and $a_1 < a_2$, show that the trend statistic equals $Z_2(\tau)/\sqrt{\hat{\sigma}_{22}} = -Z_1(\tau)/\sqrt{\hat{\sigma}_{11}}$.

Solution [Supplement].

(a) By the formulas for $\hat{\sigma}_{kk'}$, the contribution of t_j to $\hat{\Sigma}$ is $w_j\{\mathrm{diag}(\boldsymbol{\pi}_j) - \boldsymbol{\pi}_j\boldsymbol{\pi}_j'\}$. So $\mathbf{a}'\hat{\Sigma}\mathbf{a} = \sum_j w_j\{\sum_k \pi_{jk}a_k^2 - (\sum_k \pi_{jk}a_k)^2\}$. Since $\pi_{jk} \geq 0$ and $\sum_k \pi_{jk} = 1$, the brace is the variance of a_κ when the group label κ is drawn with probabilities $\boldsymbol{\pi}_j$, hence ≥ 0; also $w_j \geq 0$.

(b) A variance vanishes iff the variable is a.s. constant, i.e. iff a_k is constant on $\{k : \pi_{jk} > 0\} = \{k : Y_{jk} > 0\}$. A sum of nonnegative terms is positive iff some term is, which proves the equivalence. For strictly increasing scores, two groups at risk make the a_k unequal, and $W(t_j) \neq 0$, $1 \leq d_j < Y_j$ give $w_j > 0$.

(c) Since $Z_1 = -Z_2$ and, by Exercise 9.3(a), $\hat{\sigma}_{11} = -\hat{\sigma}_{12} = \hat{\sigma}_{22}$,

$$\mathbf{a}'\mathbf{Z} = (a_2 - a_1)Z_2, \qquad \mathbf{a}'\hat{\Sigma}\mathbf{a} = (a_2 - a_1)^2\hat{\sigma}_{22}.$$

Dividing, and using $a_2 - a_1 > 0$, gives $Z_2/\sqrt{\hat{\sigma}_{22}} = -Z_1/\sqrt{\hat{\sigma}_{11}}$: for two groups the trend test is the one-sided two-sample test of $h_1 \leq h_2$. ■

Exercise 10.2 (The paired log-rank test as an exact sign test)

In pair i ($i = 1, \ldots, M$, pairs independent) let X_{1i}, X_{2i} be i.i.d. continuous and both censored at a common time C_i independent of (X_{1i}, X_{2i}). Let D_1 (D_2) be the number of pairs in which member 1 (member 2) has an observed event first, and $m = D_1 + D_2$.

(a) Show that, conditionally on m, $D_1 \sim \mathrm{Bin}(m, 1/2)$.

(b) Show that $\mathrm{E}(D_1 - D_2 \mid m) = 0$ and $\mathrm{Var}(D_1 - D_2 \mid m) = m$, matching the paired log-rank statistic.

(c) Compute the exact two-sided p-value for the 6-MP pairs ($D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$).

Solution [Supplement].

(a)

Step 1. Member 1 has an observed event first iff $A_{1i} = \{X_{1i} < X_{2i},\ X_{1i} \le C_i\}$, and similarly A_{2i} (ties have probability 0). These events are disjoint.

Step 2. Swapping X_{1i} and X_{2i} leaves the joint law of (X_{1i}, X_{2i}, C_i) unchanged and maps A_{1i} onto A_{2i}, so $P(A_{1i}) = P(A_{2i})$.

Step 3. Let $\xi_i = 1, 2, 0$ according as A_{1i}, A_{2i}, or neither occurs. The ξ_i are independent, and $P(\xi_i = 1 \mid \xi_i \neq 0) = 1/2$ by Step 2. Conditionally on the set $J = \{i : \xi_i \neq 0\}$, the ξ_i, $i \in J$, are independent fair choices between 1 and 2, so $D_1 \mid J \sim \mathrm{Bin}(|J|, 1/2)$. This depends on J only through $|J| = m$, so the same holds given m.

(b) $D_1 - D_2 = 2D_1 - m$, so $\mathrm{E}(D_1 - D_2 \mid m) = 2 \cdot \frac{m}{2} - m = 0$ and $\mathrm{Var}(D_1 - D_2 \mid m) = 4 \cdot \frac{m}{4} = m$. With $W \equiv 1$ the paired statistic is $(D_1 - D_2)/\sqrt{D_1 + D_2}$, exactly the standardized difference.

(c) With $m = 21$ and $D_1 = 18$,

$$p = 2P\{\mathrm{Bin}(21, \tfrac{1}{2}) \ge 18\} = 2 \cdot \frac{\binom{21}{18} + \binom{21}{19} + \binom{21}{20} + \binom{21}{21}}{2^{21}} = \frac{2(1330 + 210 + 21 + 1)}{2\,097\,152} \approx 0.0015,$$

in agreement with the normal approximation $p \approx 0.001$. ■